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College of Engineering - Department of Electronics Engineering
Bachelor of Science in Mechatronics Engineering

“Item Speed Control in Conveyor Systems for consistent inventory discharge utilizing IR Proximity Sensors”

**ICE - 405 SENSORS ENGINEERING & ECE - 425 CONTROL SYSTEMS
ENGINEERING**

MExE-3202

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I. Abstract

This project presents the design, mathematical modeling, simulation, and prototype implementation of a closed-loop conveyor object spacing control system utilizing infrared proximity sensors for consistent inventory discharge. The system employs an asymmetric Proportional-Integral (PI) control strategy implemented on an Arduino Uno microcontroller, using two HW201 infrared proximity sensors, an L298N motor driver module, and a custom 3D-printed conveyor belt driven by a TT DC gear motor. The control objective is to maintain a consistent inter-item discharge interval of 900 milliseconds by dynamically adjusting conveyor belt speed in response to real-time spacing feedback detected at the entry and discharge points of the conveyor.

The mathematical foundation was developed through transfer function and state-space modeling of the electrical, mechanical, electromechanical, and gear subsystems. An asymmetric PI control law with separate braking and accelerating gain sets was implemented alongside anti-windup protection, asymmetric PWM ramping, and a feed-forward pre-compensation mechanism to mitigate the belt's inherent transport delay. A First-In First-Out queue structure preserves measurement-to-item correspondence during multi-object operation.

MATLAB and Simulink were used for system analysis and simulation, supported by an interactive digital twin for controller validation prior to hardware deployment. Prototype testing under close-spacing, far-spacing, and mixed-spacing conditions confirmed the system's ability to apply accurate bidirectional corrective actions. Results demonstrate the effectiveness of closed-loop PI control in improving conveyor discharge consistency, accuracy, and adaptability for industrial automation applications.



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II. Introduction

This project titled “Item Speed Control in Conveyor Systems for consistent inventory discharge utilizing IR Proximity Sensors” seeks to portray in small scale how closed-loop PI controllers can be utilized in conveyor systems. The project uses a custom designed 3D printed conveyor system which utilizes a DC gear motor to drive and actuate the conveyor belt. The entire logic of the system is programmed and controlled through an arduino uno along with two infrared proximity sensors that are used to detect the presence and determine the spacing of objects along the conveyor belt. By utilizing the detection signals produced by the infrared sensors, object spacing can be monitored and belt discharge can be controlled. If the said objects are too close to one another the system would slow down to prevent item congestion in a supposed station after the belt, and if objects were to have a further distance from one another than the target set point the system would speed up for the second object to compensate for delivery delay.

Engineering an automated system that integrates mechanics, electronics, and computing into a singular system is the interdisciplinary engineering field of mechatronics, however being able to control every component to work in unison to function as a singular system is only the next step into mechatronics engineering. Closed-loop control is an essential concept in mechatronics for automated systems because it allows continuous operations based on real-time feedback within the system. Unlike open-loop systems, which operate without considering the actual output condition, closed-loop systems utilize sensors to detect system behavior and provide feedback to the controller for corrective action. This enables the system to maintain accuracy, stability, and consistent performance even when disturbances or varying operating conditions are present. It is widely used in industrial automation, robotics, conveyor systems, motor speed regulation, and manufacturing processes where maintaining controlled and stable

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operation is critical. Through continuous feedback and adjustment, closed-loop control helps minimize errors, reduce manual intervention, improve safety, and optimize overall system performance. In the proposed conveyor speed control system, applying a closed-loop control plays an essential role in this system for monitoring and regulating inventory speed along the conveyor belt for maintaining a consistent inventory discharge rate. By applying closed-loop control in this project, the project design achieves being an automated mechatronic system which improves efficiency, precision, and reliability by allowing machines to adapt automatically to change within the system. Unlike traditional conveyor systems with no such application of logic controllers, this system obtains its feedback for its closed loop system from the infra-red proximity sensors allowing the controller to dynamically adjust the conveyor speed in response to changing conditions. This adaptive behavior demonstrates the practical importance of closed-loop control in achieving efficient and reliable automated material handling operations

OBJECTIVES:

1. To design a proportional control system for conveyor belts that maintains optimal material flow by adjusting speed based on inventory item flow, and its corresponding mathematical control model.
2. To develop a belt control system that implements the functionality of an IR sensor module to regulate item flow for consistent discharge.
3. To analyze and evaluate the system's performance in terms of the following
 - Speed response
 - Control accuracy
 - Sensor reliability



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III. System Description

HARDWARE CONFIGURATION

The system is tailored to showcase the application of a closed-loop feedback architecture in automated conveyor systems. The overall mechanical hardware design is structured so the project is suited for safe DC power. The primary power supply supplies approximately 13.7V that is then regulated through a LM2596 DC-DC step-down buck converter. To improve the electrical safety of the electrical components in this project, a toggle switch is used as the main power switch, so current doesn't flow immediately after plugging in the power supply. Two limit switch push buttons are then utilized to create a start and stop function in the project, this allows for confirmation that all components are functional before manually starting the system and when observation is done the system can then be manually and safely switched off without aggressively unplugging the power, this also allows the operator to manually switch off the project if any errors were to occur and then resetting via the start button. Further hardware configuration can be observed in *figure 2.1*.



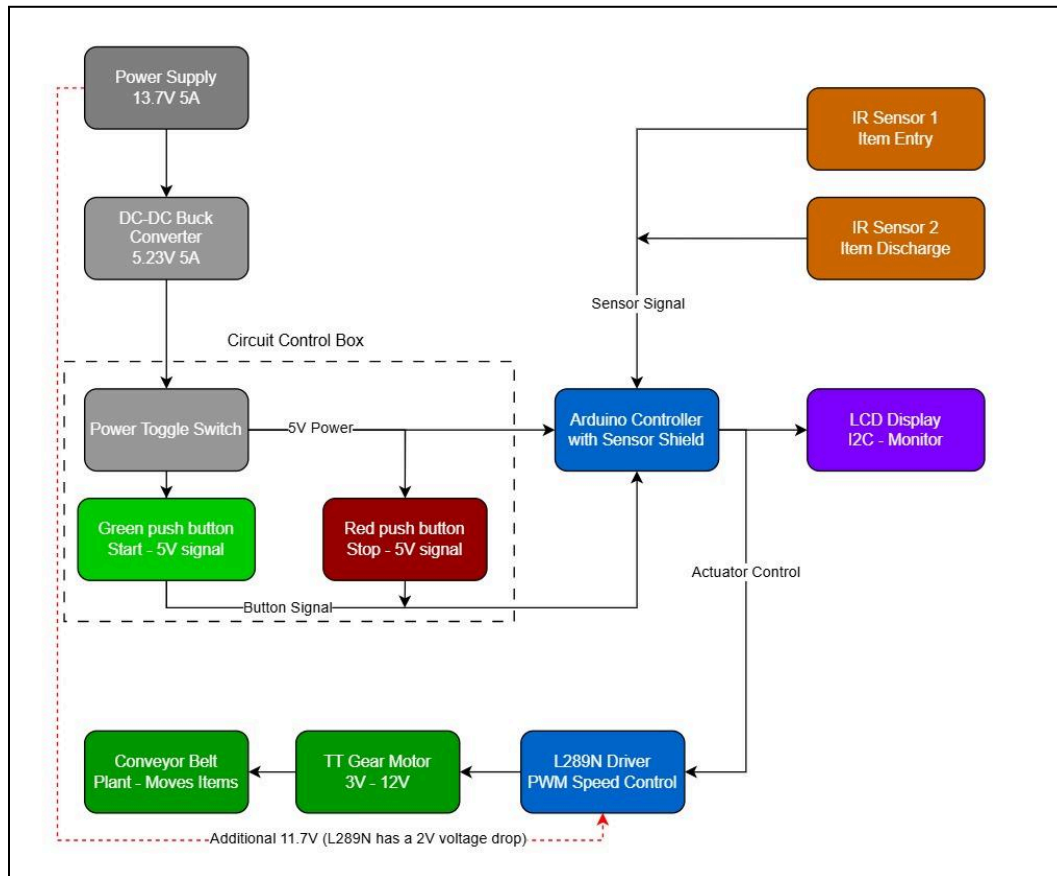
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“figure 2.1. (Hardware Configuration)”

SENSOR

The sensor to be used in this system is a HW201 Infrared (IR) Sensor. This sensor measures and detects the infrared radiation which is emitted from the TR transmitter and when it hits an object the radiation would then be reflected into the IR receiver. This process is then utilized to create a signal output. When no radiation is reflected, the electrical resistance is increased, then if radiation is reflected, the resistance will decrease (resistance will vary on the distance it was reflected from).



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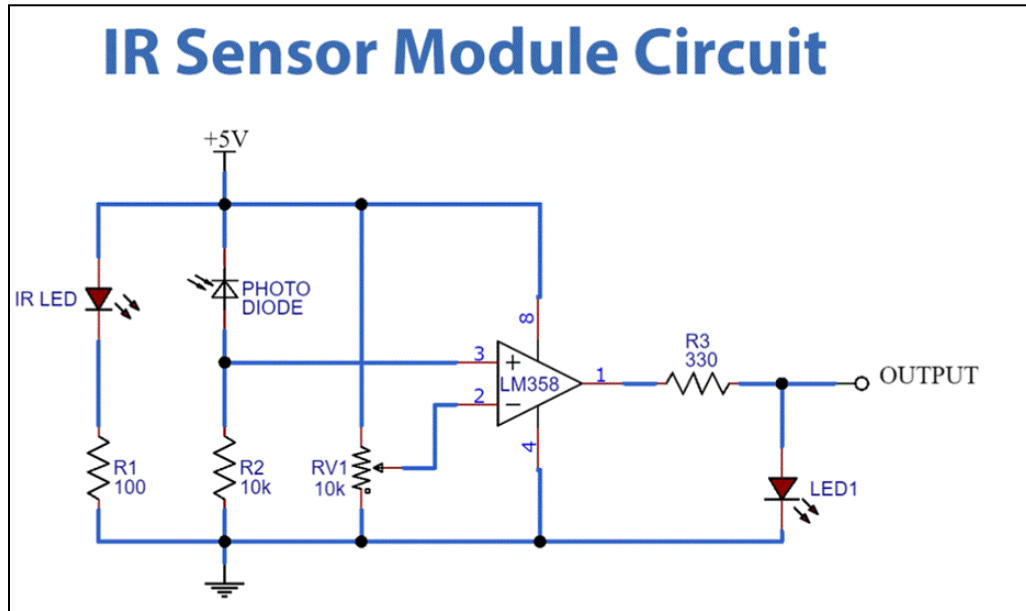
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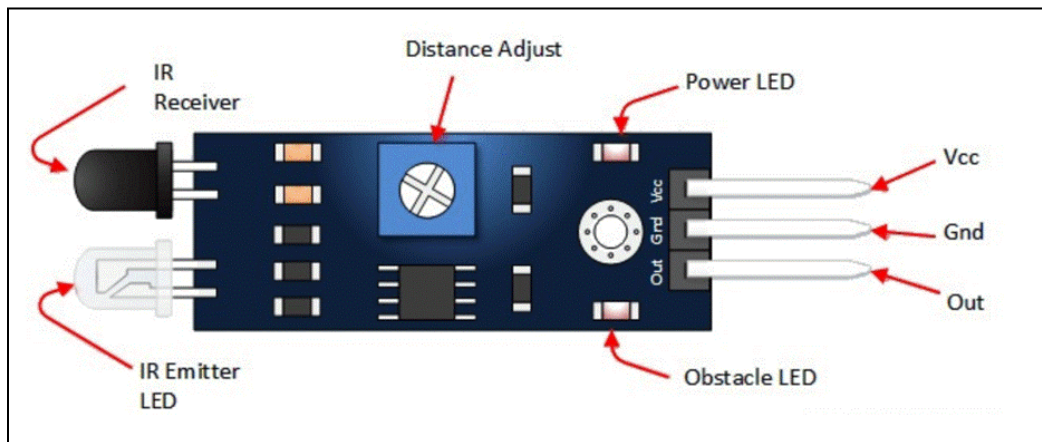
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SENSOR DATASHEET



“figure 2.2. (Circuit Configuration of an IR Sensor)”



“figure 2.3. (IR Sensor)”



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Pinouts of HW201 Module	
PIN NAME	DESCRIPTION
VCC	Power Supply Input
GND	Power Supply Ground
OUT/SIGNAL	Active High Output

SPECIFICATIONS

- Main Chip: LM393
- Detection Distance: 2~30cm
- Detection Angle: 35 °
- Working Voltage: 3.3V~5V
- Board Size: 31 * 14mm / 1.22 * 0.55in
- Board Weight(1pc): 3g
- 20mA supply current
- 2 to 30cm adjustable sensing range (on-board Potentiometer)

SENSOR CONDITIONING

The HW201 infrared (IR) sensor module utilizes an LM358 operational amplifier to perform signal conditioning on the output of the IR receiver. When infrared light emitted by the transmitter is reflected by a nearby object and detected by the photodiode (or phototransistor), it generates a small analog voltage proportional to the reflected intensity. However, this signal is typically weak and susceptible to noise, making it unsuitable for direct use in control systems.



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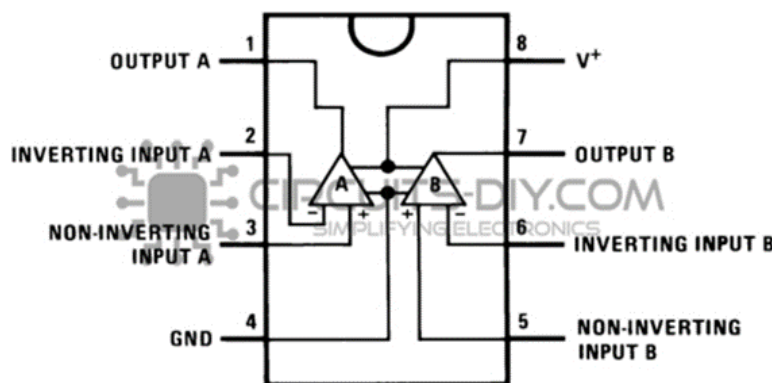
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To address this, the LM358 op-amp is configured as a comparator with adjustable threshold. One input of the LM358 receives the analog signal from the IR receiver, while the other input is connected to a reference voltage set by an onboard potentiometer. The amplifier compares these two voltages and produces a clean digital output: when the sensor signal exceeds the threshold, the output switches state (HIGH or LOW), indicating object detection.

This signal conditioning process improves reliability by amplifying the weak sensor signal, filtering out minor fluctuations, and converting it into a stable, noise-resistant digital signal that can be easily interfaced with microcontrollers such as an Arduino or PLC systems.

LM358 Pinout

LM358 Pinout



“figure 2.4. (IR Sensor Op-Amp)”



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IMPLEMENTATION OF THE SENSOR IN THE SYSTEM

To manage the flow of inventory items within the system, an IR proximity sensor is implemented to detect discrete events corresponding to item entry onto the conveyor belt. Each detection provides a measurable signal indicating the presence of an item, enabling the system to track the timing and frequency of incoming materials. This event-based sensing approach allows the system to estimate the infeed rate and monitor consistency in item flow. The acquired data is then used as feedback for adjusting conveyor speed to maintain a stable discharge rate. By integrating discrete sensing with control logic via an Arduino Uno, the system achieves more reliable and responsive inventory handling.

ELECTRICAL DESIGN

An arduino uno R3 is used as this project's main embedded microcontroller to handle the project's logic and control calculations. The I2C ports are used so the operator can monitor the real time speed, gap, item count, and error in the system. The DC-DC buck converter ensures that approximately 5.23V is used to power the electrical components and the 13.7V from the power supply is used as additional power for the TT gear motor which has an operating voltage of 3-12V, in this set up 13.7V is used because as the additional voltage flows through the motor driver module L298N there is a 2V voltage drop. Then two infra-red proximity sensors are connected to the arduino board, because the system only requires the sensors to be used to detect objects entering and leaving the belt, the sensors deliver discrete signals to the digital GPIO's.



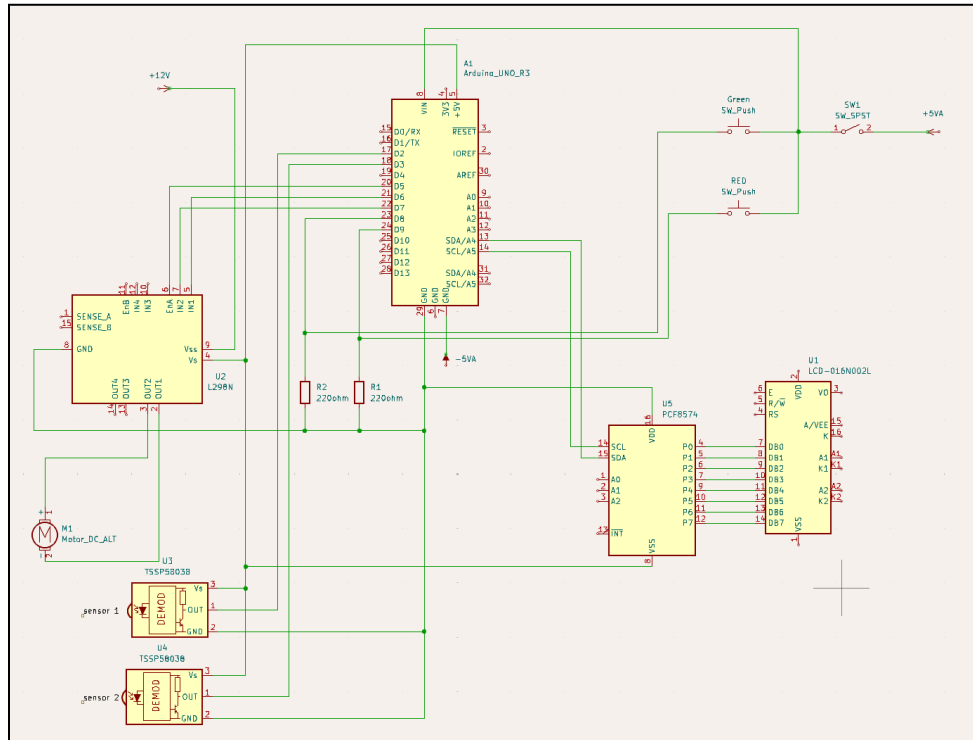
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“figure 2.5. (Electrical Schematic)”

CAD DESIGN AND MANUFACTURING

The project was then designed on Solidworks to have a conceptual design before physically assembling it. The system features a custom made conveyor belt which utilizes a custom made timing belt design. The belt length is 368mm from the gear to gear centers. The custom parts were then 3D printed with PETG, the total mass used for the prints is 527.73g. The parts manufactured for this project consists of an inflow ramp which has a 30 degree angle of tilt and a mounting slot for the infra-red proximity sensor. A vertical elevation support for the LCD display which allows the operator to view the real-time system conditions at a comfortable viewing angle. For the system discharge a basket is placed to receive items that will be discharged from the conveyor system. screw supports were made for the conveyor belt to hold it

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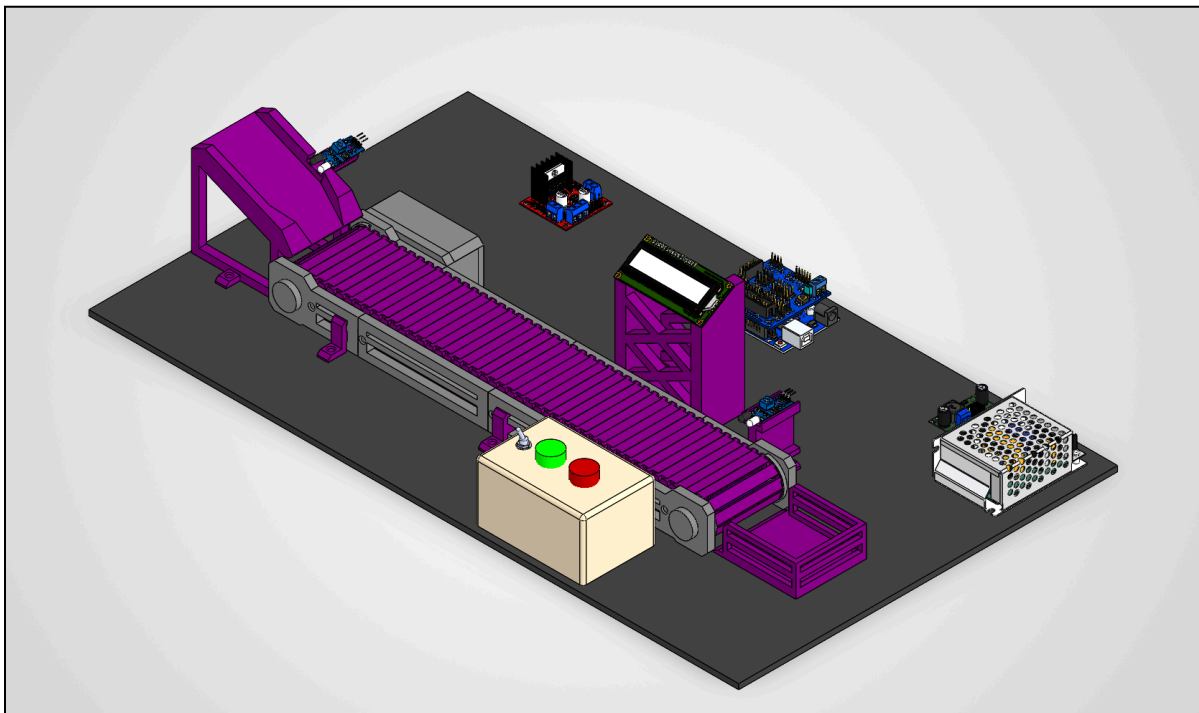
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in place while retaining the modular ability of the conveyor if modifications are needed. As for the conveyor itself, a motor chassis was made to hold the DC motor in place and to protect it from external disturbances. Two gears are present in the conveyor system, one being the driver pulley and the other being the driven pulley. The frame of the conveyor system is designed to hold the belt in a fixed position to prevent belt slack without the need of additional tension. Finally for the belt, it uses a timing belt structure to allow it to be driven by the pulleys by having both rigid bodies interlock with each other. To achieve the timing belt design 98 individual pieces were printed using a print in place technique with a 0.5mm tolerance to allow for optimal flexibility of the belt for the conveyor system.



“figure 2.6. (CAD Isometric view)”



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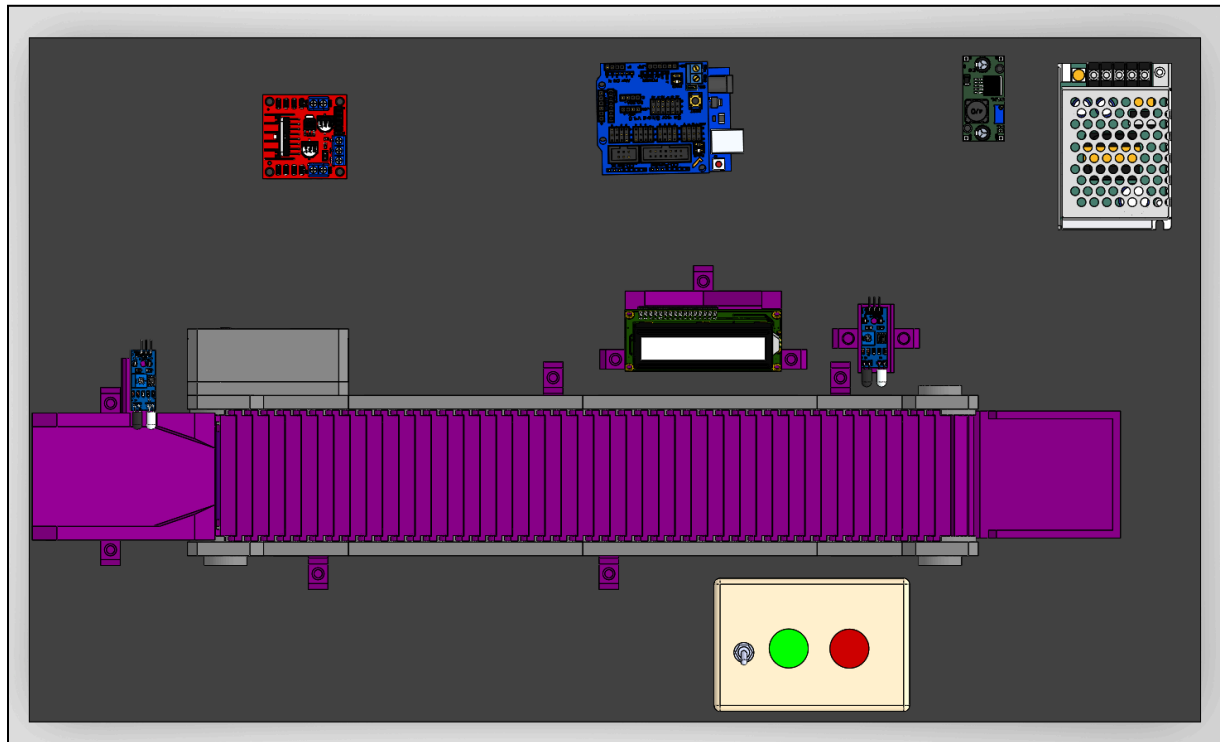
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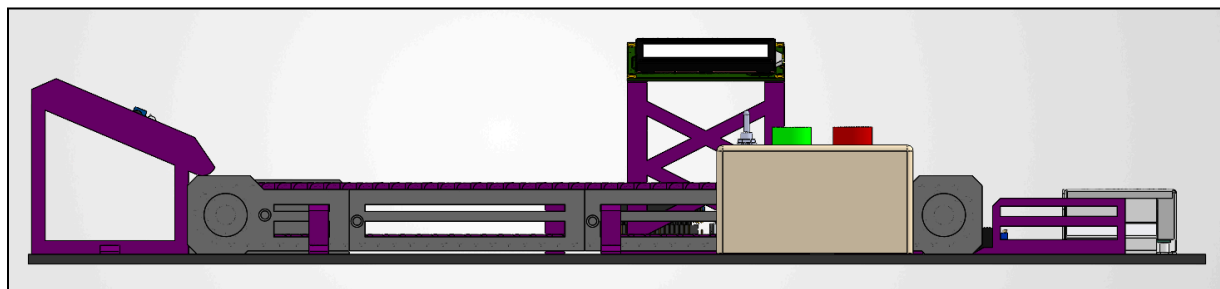
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“figure 2.7. (CAD Top view)”



“figure 2.8. (CAD Front view)”



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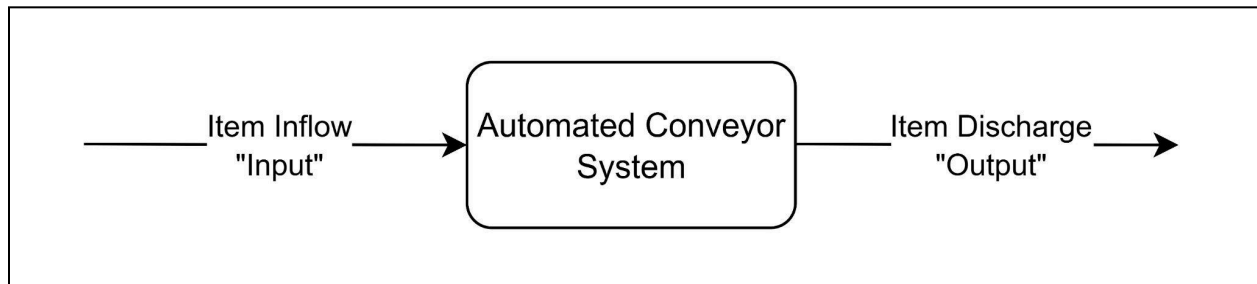
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CONTROL SYSTEM

For this automated conveyor system where speed control is the main variable to be controlled, *figure 2.9.* provides a basic input output diagram of how the system works overall. where items entering the conveyor serves as the input then the conveyor system regulates the speed ensuring that the items on the belt are discharged consistently. The items that are discharged from the conveyor belt would also be the output of the system.



“figure 2.9. (Basic input output diagram)”

In this automated conveyor system, maintaining a consistent discharge rate is essential for material handling in automated systems especially in industrial applications. For the control law to be applied to this project its important to choose the most appropriate PID controller was considered, however it was deemed overly complex for this system since the main priority is creating a stable system response and reducing steady state errors, there is no need for the system to predict errors also this effects the arduino computational load hindering the response time of the system.



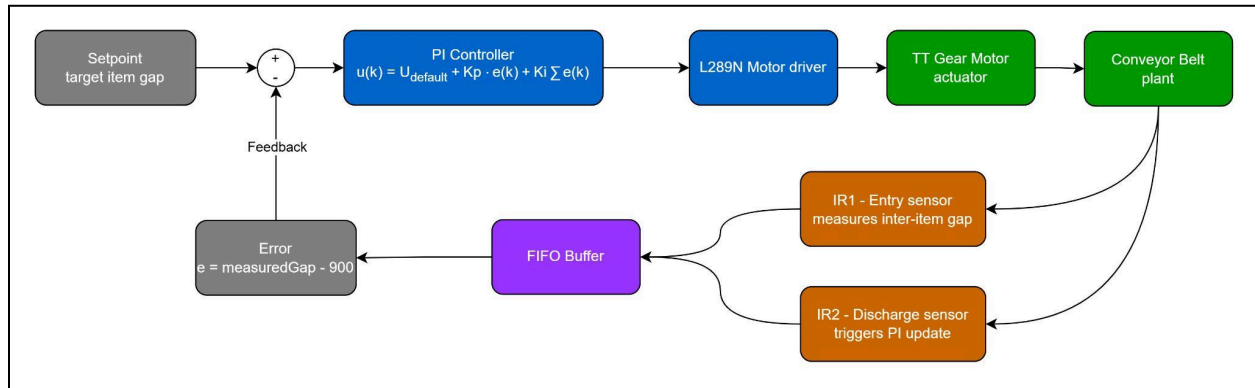
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“figure 2.10. (Block diagram of the closed-loop system)”

Controller Type

PI Control law (Asymmetric Gains)

$$u(k) = U_{\text{default}} + K_p \cdot e(k) + K_i \cdot \sum e$$

With gain selection based on error sign:

Condition	Meaning	K_p	K_i
$e(k) < 0$	Items too close → slow down	0.18	0.005
$e(k) \geq 0$	Items too far → speed up	0.07	0.002

$U_{\text{default}} = 110$ (PWM), with output clamped: $90 \leq u(k) \leq 200$



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Anti-Windup (Conditional Integration)

The integral only accumulates when the output is not saturated in the direction of the error:

integrate if:

$$\text{NOT} [(e < 0 \text{ AND } u \leq U_{\min}) \text{ OR } (e > 0 \text{ AND } u \geq U_{\max})]$$

Integral is also hard-clamped: $-1000 \leq \sum e \leq 1000$, limiting the maximum integral contribution to $\pm 5\text{PWM}$ units.

Asymmetric PWM Ramp (Motor Actuator)

$$\text{PWM}_{\text{current}}(t) = \begin{cases} \text{PWM}_{\text{current}} - 5 & \text{if } \text{PWM}_{\text{current}} > \text{targetPWM} \\ \text{PWM}_{\text{current}} + 2 & \text{if } \text{PWM}_{\text{current}} < \text{targetPWM} \end{cases}$$

Deceleration is $2.5\times$ faster than acceleration — prevents item pile-up on sudden slowdown.

Feed-Forward Trigger Condition

$$\text{IF } \Delta t_{\text{entry}} < 700 \text{ ms} \Rightarrow \text{schedule FF event at } t + 1000 \text{ ms, } u_{\text{FF}} = 120 \text{ PWM}$$

This is a pre-emptive, one-shot speed reduction that fires midway through the belt's ~ 2 s transport delay. IR2's PI output always cancels and overrides it upon item exit.

Inter-Item Gap Measurement (at IR1)

$$\Delta t_{\text{entry}}(k) = t_{\text{now}} - t_{\text{prev_entry}}$$



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Measured only for intra-batch items (when before the new arrival) to prevent phantom cross-batch gaps from entering the FIFO queue.

Error Signal

$$e(k) = \text{measuredGap}(k) - \text{TARGET_GAP}$$

Where TARGET_GAP = 900 ms. This is computed event-driven — once per item exit at IR2, not every loop tick.

The overall architecture is a feedback PI controller with a feed-forward pre-compensation path. The PI loop closes at IR2 (exit), while the feed-forward path acts at IR1 (entry) to reduce the impact of the ~2-second transport delay, a classic Smith Predictor-inspired approach adapted for discrete, event-driven control.



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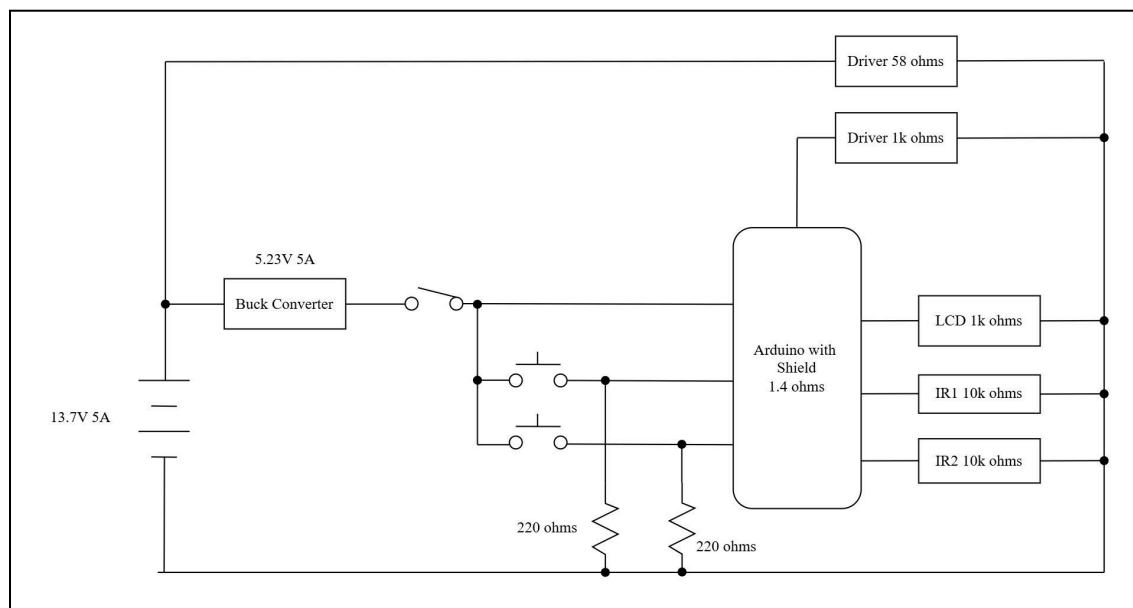
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IV. Mathematical Modeling

4.1 Transfer Function Modeling

ELECTRICAL SYSTEM

The mathematical model for this project's electrical system is derived from its own electrical circuit. To convert the enter circuit into a RLC circuit for the mathematical modelling, all modules used in this circuit are assumed to be load resistors, the resistance of each component was measured from their respective VCCs to GNDs. For the arduino board, the resistance is measured from its VCC to its Vout where the voltage would flow to supply the respective circuit components. The motor driver module exhibits two voltage ins and a shared ground, two load resistors are then assumed from this module, one that connects the additional voltage directly from the power supply and one that attains voltage directly from the arduino board.



“figure 4.1. Electrical System”



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Loop 1

$$v(t) = R_{Arduino} \cdot i_T(t) + R_1 \cdot i_1(t)$$

$$v(t) = 1.4 \cdot i_T(t) + 58 \cdot i_1(t)$$

$$V(s) = 1.4 \cdot I_T(s) + 58 \cdot I_1(s)$$

Loop 2

$$v(t) = R_{Arduino} \cdot i_T(t) + R_2 \cdot i_2(t)$$

$$v(t) = 1.4 \cdot i_T(t) + 1000 \cdot i_2(t)$$

$$V(s) = 1.4 \cdot I_T(s) + 1000 \cdot I_2(s)$$

Loop 3

$$v(t) = R_{Arduino} \cdot i_T(t) + R_3 \cdot i_3(t)$$

$$v(t) = 1.4 \cdot i_T(t) + 1000 \cdot i_3(t)$$

$$V(s) = 1.4 \cdot I_T(s) + 1000 \cdot I_3(s)$$

Loop 4

$$v(t) = R_{Arduino} \cdot i_T(t) + R_4 \cdot i_4(t)$$

$$v(t) = 1.4 \cdot i_T(t) + 10000 \cdot i_4(t)$$



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$$V(s) = 1.4 \cdot I_T(s) + 10000 \cdot I_4(s)$$

Loop 5

$$v(t) = R_{Arduino} \cdot i_T(t) + R_5 \cdot i_5(t)$$

$$v(t) = 1.4 \cdot i_T(t) + 10000 \cdot i_5(t)$$

$$V(s) = 1.4 \cdot I_T(s) + 10000 \cdot I_5(s)$$

Loop 6

$$v(t) = R_{220} \cdot i_6(t)$$

$$R_{220,eq} = \frac{220 \times 220}{220 + 220} = \frac{48400}{440} = 110 \, \Omega$$

$$v(t) = 110 \cdot i_6(t)$$

$$V(s) = 110 \cdot I_6(s)$$

$$I_T(s) = I_1(s) + I_2(s) + I_3(s) + I_4(s) + I_5(s)$$

$$I_T(s) = V_{load}(s) \left(\frac{1}{58} + \frac{1}{1000} + \frac{1}{1000} + \frac{1}{10000} + \frac{1}{10000} \right)$$



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$$I_T(s) = V_{load}(s)(0.01724 + 0.001 + 0.001 + 0.0001 + 0.0001)$$

$$I_T(s) = V_{load}(s)(0.01944)$$

$$I_T(s) = \frac{V_{load}(s)}{R_{parallel}}$$

$$R_{parallel} = \frac{1}{0.01944} = 51.44 \, \Omega$$

$$I(s) = I_T(s) + I_6(s)$$

$$I(s) = I_T(s) + I_6(s) = \frac{V(s)}{52.84} + \frac{V(s)}{110}$$

$$I(s) = V(s) \left(\frac{1}{52.84} + \frac{1}{110} \right)$$

$$I(s) = V(s)(0.01892 + 0.00909)$$

$$I(s) = V(s)(0.02801)$$



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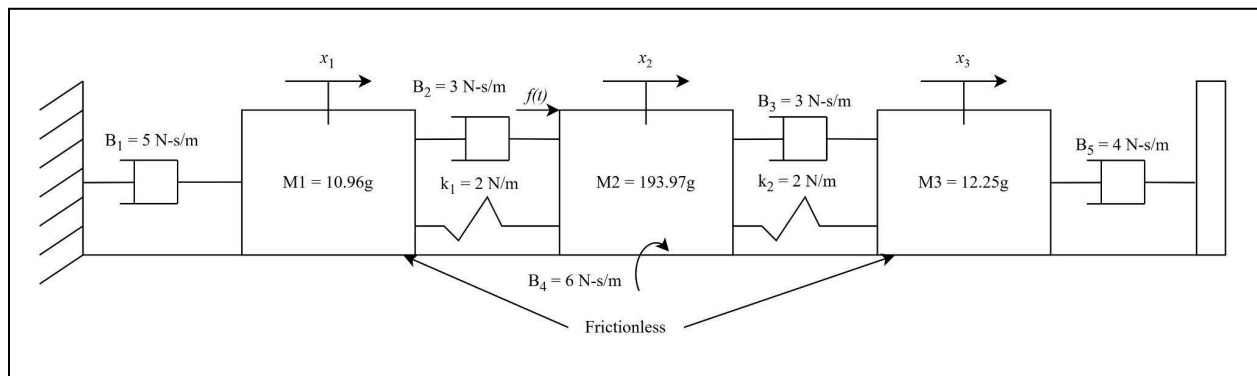
$$I(s) = \frac{V(s)}{35.70}$$

Final Transfer Function

$$\frac{I(s)}{V(s)} = \frac{1}{35.70}$$

TRANSLATIONAL MECHANICAL SYSTEM

something something technical statements



“figure 4.2. Translational System”



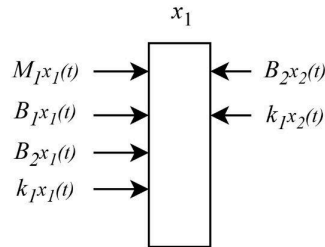
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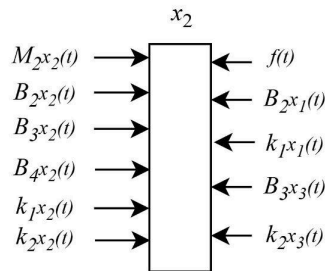
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$$M_1 s^2 X_1 + B_1 s X_1 + B_2 s (X_1 - X_2) + k_1 (X_1 - X_2) = 0$$

$$X_1 (M_1 s^2 + (B_1 + B_2)s + k_1) - X_2 (B_2 s + k_1) = 0$$

$$X_1 (10.96s^2 + 8s + 2) - X_2 (3s + 2) = 0$$



$$M_2 s^2 X_2 + (B_2 + B_3 + B_4)s X_2 + k_1 X_2 + k_2 X_2 - B_2 s X_1 - k_1 X_1 - B_3 s X_3 - k_2 X_3 = F(s)$$

$$-X_1 (B_2 s + k_1) + X_2 (M_2 s^2 + (B_2 + B_3 + B_4)s + k_1 + k_2) - X_3 (B_3 s + k_2) = F(s)$$

$$-X_1 (3s + 2) + X_2 (193.97s^2 + 12s + 4) - X_3 (3s + 2) = F(s)$$



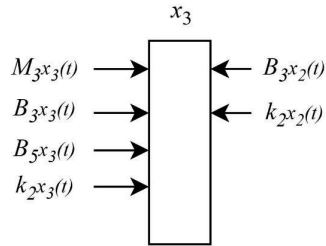
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$$M_3 s^2 X_3 + (B_3 + B_5) s X_3 + k_2 (X_3 - X_2) - B_3 s X_2 = 0$$

$$-X_2 (B_3 s + k_2) + X_3 (M_3 s^2 + (B_3 + B_5) s + k_2) = 0$$

$$-X_2 (3s + 2) + X_3 (12.25s^2 + 7s + 2) = 0$$

$$\begin{bmatrix} 10.96s^2 + 8s + 2 & -(3s + 2) & 0 \\ -(3s + 2) & 193.97s^2 + 12s + 4 & -(3s + 2) \\ 0 & -(3s + 2) & 12.25s^2 + 7s + 2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} = \begin{bmatrix} 0 \\ F(s) \\ 0 \end{bmatrix}$$

$$\Delta = a_1 \begin{bmatrix} b_2 & c_2 \\ b_3 & c_3 \end{bmatrix} - b_1 \begin{bmatrix} a_2 & c_2 \\ a_3 & c_3 \end{bmatrix} + c_1 \begin{bmatrix} a_2 & b_2 \\ a_3 & b_3 \end{bmatrix}$$

$$\Delta = a_1 (b_2 \cdot c_3 - c_2 \cdot b_3) - b_1 (a_2 \cdot c_3 - c_2 \cdot a_3) + c_1 (a_2 \cdot b_3 - b_2 \cdot a_3)$$



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$$\Delta = a_1(b_2c_3 - c_2b_3) - b_1(a_2c_3 - c_2a_3)$$

$$b_2c_3 - c_2b_3 = (193.97s^2 + 12s + 4)(12.25s^2 + 7s + 2) - (-(3s + 2))(-(3s + 2))$$

$$= (193.97s^2 + 12s + 4)(12.25s^2 + 7s + 2) - (3s + 2)^2$$

$$a_2c_3 - c_2a_3 = (-(3s + 2))(12.25s^2 + 7s + 2) - (-(3s + 2))(0)$$

$$= -(3s + 2)(12.25s^2 + 7s + 2)$$

$$\Delta = a_1[(193.97s^2 + 12s + 4)(12.25s^2 + 7s + 2) - (3s + 2)^2] - b_1[-(3s + 2)(12.25s^2 + 7s + 2)]$$

$$\Delta = (10.96s^2 + 8s + 2)[(193.97s^2 + 12s + 4)(12.25s^2 + 7s + 2) - (3s + 2)^2] - (3s + 2)^2(12.25s^2 + 7s + 2)$$

$$N_{X_2} = \begin{bmatrix} a_1 & 0 & c_1 \\ a_2 & F(s) & c_2 \\ a_3 & 0 & c_3 \end{bmatrix}$$

$$N_{X_2} = F(s)(-1)^{2+2} \begin{bmatrix} a_1 & c_1 \\ a_3 & c_3 \end{bmatrix}$$

$$= F(s)(a_1c_3 - c_1a_3)$$

$$= F(s)(a_1c_3 - 0)$$

$$= F(s) \cdot a_1c_3$$



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$$N_{X_2} = F(s)(10.96s^2 + 8s + 2)(12.25s^2 + 7s + 2)$$

$$\frac{X_2(s)}{F(s)} =$$

$$\frac{(10.96s^2 + 8s + 2)(12.25s^2 + 7s + 2)}{(10.96s^2 + 8s + 2)[(193.97s^2 + 12s + 4)(12.25s^2 + 7s + 2) - (3s + 2)^2] - (3s + 2)^2(12.25s^2 + 7s + 2)}$$

Final Transfer Function

$$\frac{X_2(s)}{F(s)} = \frac{134.26s^4 + 174.72s^3 + 102.42s^2 + 30s + 4}{s(26042.49s^5 + 35501.62s^4 + 22291.22s^3 + 7333.42s^2 + 1236.72s + 60)}$$

ROTATIONAL MECHANICAL SYSTEM

The rotational system of the project mainly focuses on the main motor shaft along with the driver pulley and driven pulley of the conveyor belt. J1 represents the motor shaft that is connected in series with the driver pulley, their connection experiences friction along the conveyor frame and the connection between two rigid bodies also experiences torsional stiffness. A spring and damper are used to connect J2 with J3 which also represents the belt. The dampers b2, b3, and b4 represent the friction the pulleys experience in connection with the conveyor frame.



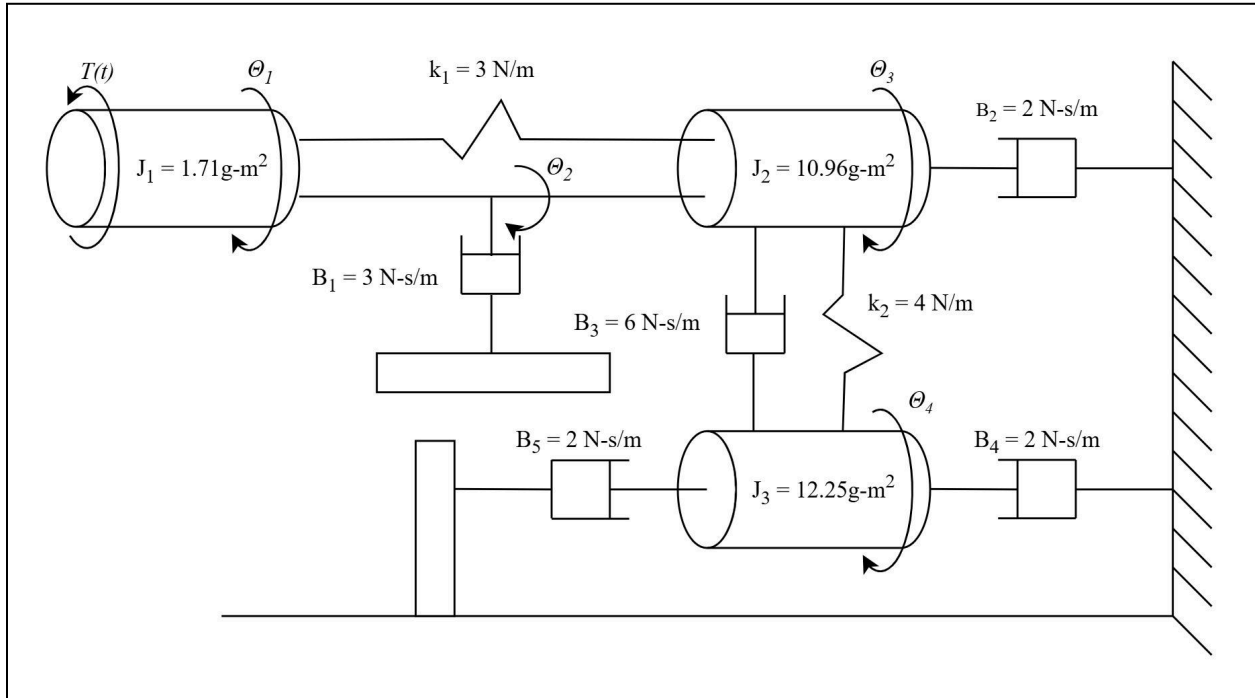
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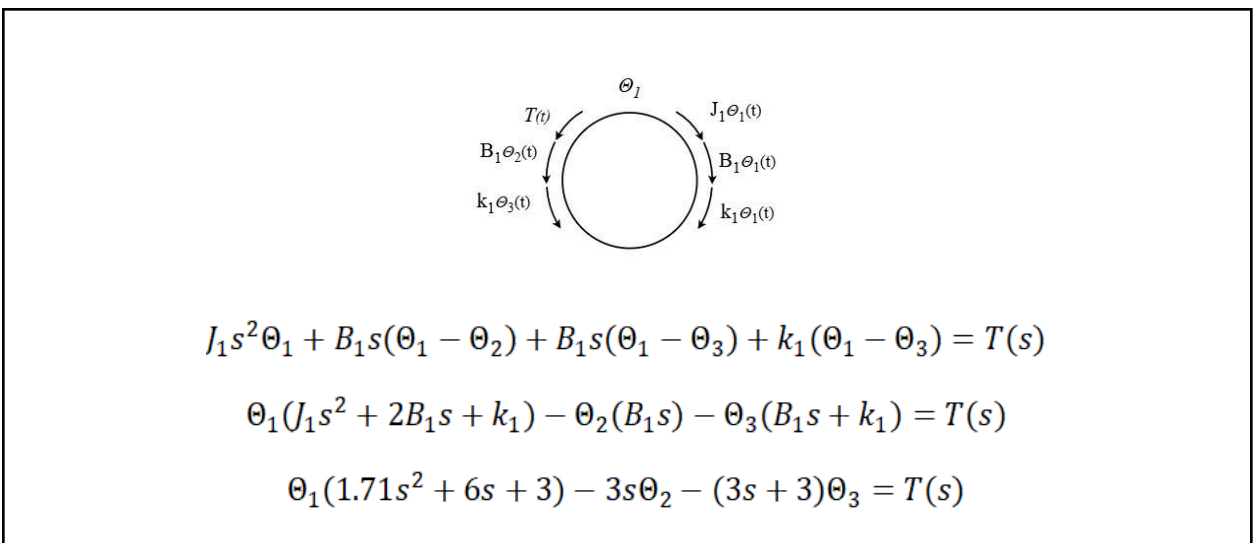
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“figure 4.3. Rotational System”





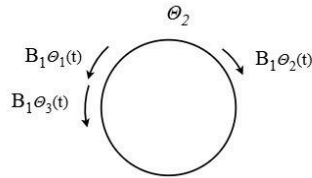
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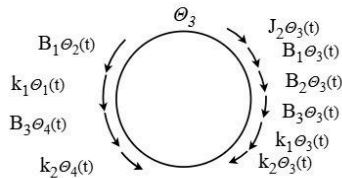
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$$B_1 s(\Theta_2 - \Theta_1) + B_1 s(\Theta_2 - \Theta_3) = 0$$

$$-B_1 s\Theta_1 + 2B_1 s\Theta_2 - B_1 s\Theta_3 = 0$$

$$-3s\Theta_1 + 6s\Theta_2 - 3s\Theta_3 = 0$$



$$J_2 s^2 \Theta_3 + (B_3 + B_5) s \Theta_3 + k_1 \Theta_3 + k_2 \Theta_3 - B_1 s \Theta_2 - k_1 \Theta_1 = 0$$

$$-k_1 \Theta_1 - B_1 s \Theta_2 + \Theta_3 (J_2 s^2 + (B_3 + B_5) s + k_1 + k_2) = 0$$

$$-(3)\Theta_1 - 3s\Theta_2 + \Theta_3(10.96s^2 + 8s + 7) = 0$$



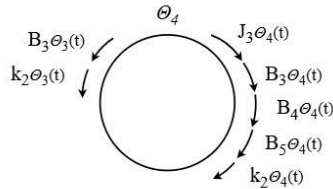
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$$J_3 s^2 \Theta_4 + (B_4 + B_2 + B_5) s \Theta_4 + k_2 \Theta_4 - B_5 s \Theta_3 - k_2 \Theta_3 = 0$$

$$-\Theta_3 (B_5 s + k_2) + \Theta_4 (J_3 s^2 + (B_2 + B_4 + B_5) s + k_2) = 0$$

$$-(2s + 4) \Theta_3 + \Theta_4 (12.25 s^2 + 6s + 4) = 0$$

$$\begin{bmatrix} 1.71s^2 + 6s + 3 & -3s & -(3s + 3) & 0 \\ -3s & 6s & -3s & 0 \\ -3 & -3s & 10.96s^2 + 8s + 7 & 0 \\ 0 & 0 & -(2s + 4) & 12.25s^2 + 6s + 4 \end{bmatrix} \begin{bmatrix} \Theta_1 \\ \Theta_2 \\ \Theta_3 \\ \Theta_4 \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta = a_1 \begin{bmatrix} b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \\ b_4 & c_4 & d_4 \end{bmatrix} - b_1 \begin{bmatrix} a_2 & c_2 & d_2 \\ a_3 & c_3 & d_3 \\ a_4 & c_4 & d_4 \end{bmatrix} + c_1 \begin{bmatrix} a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \\ a_4 & b_4 & d_4 \end{bmatrix} - d_1 \begin{bmatrix} a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \end{bmatrix}$$

$$\Delta = a_1 \begin{bmatrix} b_2 & c_2 & 0 \\ b_3 & c_3 & 0 \\ 0 & c_4 & d_4 \end{bmatrix} - b_1 \begin{bmatrix} a_2 & c_2 & 0 \\ a_3 & c_3 & 0 \\ 0 & c_4 & d_4 \end{bmatrix} + c_1 \begin{bmatrix} a_2 & b_2 & 0 \\ a_3 & b_3 & 0 \\ 0 & 0 & d_4 \end{bmatrix}$$



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$$\Delta = a_1(b_2(c_3d_4 - 0) - c_2(b_3d_4 - 0) + 0(b_3c_4 - 0))$$

$$-b_1(a_2(c_3d_4 - 0) - c_2(a_3d_4 - 0) + 0(a_3c_4 - 0))$$

$$+c_1(a_2(b_3d_4 - 0) - b_2(a_3d_4 - 0) + 0)$$

$$a_1(b_2c_3 - c_2b_3) = (1.71s^2 + 6s + 3)[(6s)(10.96s^2 + 8s + 7) - (-3s)(-3s)]$$

$$= (1.71s^2 + 6s + 3)[65.76s^3 + 48s^2 + 42s - 9s^2]$$

$$= (1.71s^2 + 6s + 3)(65.76s^3 + 39s^2 + 42s)$$

$$-b_1(a_2c_3 - c_2a_3) = -(-3s)[(-3s)(10.96s^2 + 8s + 7) - (-3s)(-3)]$$

$$= 3s[-32.88s^3 - 24s^2 - 21s - 9s]$$

$$= 3s(-32.88s^3 - 24s^2 - 30s)$$

$$= -98.64s^4 - 72s^3 - 90s^2$$

$$c_1(a_2b_3 - b_2a_3) = -(3s + 3)[(-3s)(-3s) - (6s)(-3)]$$

$$= -(3s + 3)[9s^2 + 18s]$$

$$= -(3s + 3)(9s)(s + 2)$$

$$= -9s(3s + 3)(s + 2)$$

$$= -9s \cdot 3(s + 1)(s + 2)$$



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$$= -27s(s+1)(s+2)$$

$$= -27s(s^2 + 3s + 2)$$

$$= -27s^3 - 81s^2 - 54s$$

$$\Delta = d_4[(1.71s^2 + 6s + 3)(65.76s^3 + 39s^2 + 42s) + (-98.64s^4 - 72s^3 - 90s^2) + (-27s^3 - 81s^2 - 54s)]$$

$$N_{\Theta_4} = \begin{bmatrix} a_1 & b_1 & c_1 & T(s) \\ a_2 & b_2 & c_2 & 0 \\ a_3 & b_3 & c_3 & 0 \\ 0 & 0 & c_4 & 0 \end{bmatrix}$$

$$N_{\Theta_4} = T(s)(-1)^{1+4} \begin{bmatrix} a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ 0 & 0 & c_4 \end{bmatrix}$$

$$= -T(s) \begin{bmatrix} a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ 0 & 0 & c_4 \end{bmatrix}$$

$$= -T(s)[c_4(a_2b_3 - b_2a_3)]$$

$$a_2b_3 - b_2a_3 = (-3s)(-3s) - (6s)(-3) = 9s^2 + 18s = 9s(s+2)$$

$$N_{\Theta_4} = -T(s)[-(2s+4)(9s(s+2))]$$



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$$= T(s)(2s + 4)(9s)(s + 2)$$

$$= T(s) \cdot 2(s + 2) \cdot 9s(s + 2)$$

$$= 18sT(s)(s + 2)^2$$

$$\frac{\Theta_4(s)}{T(s)} =$$

$$\frac{18s(s + 2)^2}{(12.25s^2 + 6s + 4)[(1.71s^2 + 6s + 3)(65.76s^3 + 39s^2 + 42s) - 98.64s^4 - 99s^3 - 171s^2 - 54s]}$$

$$\frac{\Theta_4(s)}{T(s)} = \frac{18s^3 + 72s^2 + 72s}{1377.51s^7 + 5116.67s^6 + 7575.69s^5 + 6300.54s^4 + 3686.40s^3 + 1224s^2 + 288s}$$

Final Transfer Function

$$\frac{\Theta_4(s)}{T(s)} = \frac{18s^2 + 72s + 72}{1377.51s^6 + 5116.67s^5 + 7575.69s^4 + 6300.54s^3 + 3686.40s^2 + 1224s + 288}$$

ELECTROMECHANICAL SYSTEM



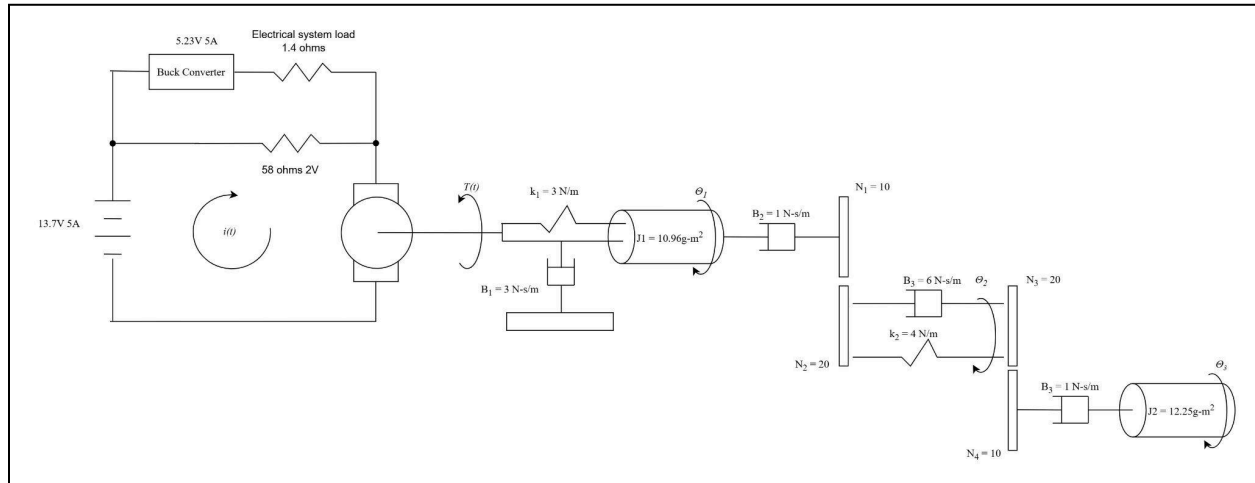
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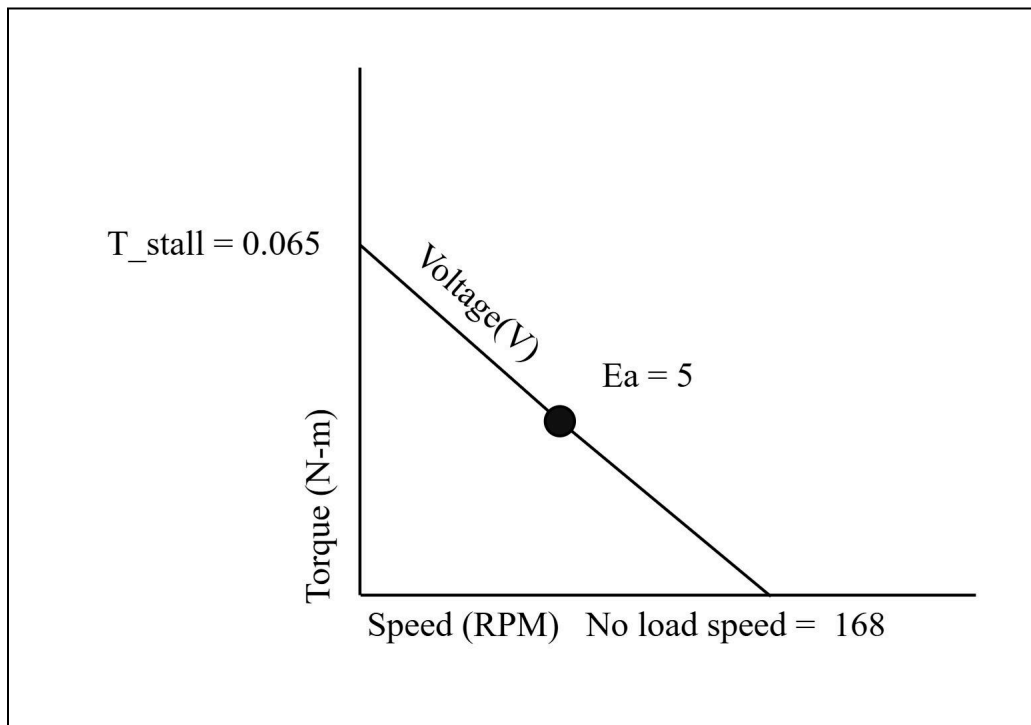
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“figure 4.4. Electromechanical System”





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$$\frac{K_t}{R_a} = \frac{T_{stall}}{E_a} = \frac{0.065}{5} = 0.013$$

$$K_b = \frac{E_a}{\omega_{no-load}} = \frac{5}{168 \times \frac{2\pi}{60}} = \frac{5}{17.59} = 0.2843$$

$$\frac{N_1}{N_2} = \frac{10}{20} = \frac{1}{2}$$

$$\frac{N_3}{N_4} = \frac{20}{10} = 2$$

$$\frac{N_1}{N_2} \times \frac{N_3}{N_4} = \frac{1}{2} \times 2 = 1$$

$$J_m = J_{motor} + J_{load} \left(\frac{N_1}{N_2} \right)^2$$

$$J_m = J_1 + J_2 \left(\frac{N_1 N_3}{N_2 N_4} \right)^2 = 10.96 + 12.25(1)^2 = 23.21$$

$$D_m = 3 + 1 + 1.5 + 1 = 6.5$$

$$\frac{\theta_m(s)}{E_a(s)} = \frac{K_t / (R_a J_m)}{s \left[s + \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \right]}$$



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$$\frac{K_t}{R_a J_m} = \frac{0.013}{23.21} = 0.0005601$$

$$\begin{aligned} & \frac{1}{J_m} \left(D_m + \frac{K_t K_b}{R_a} \right) \\ &= \frac{1}{23.21} (6.5 + 0.013 \times 0.2843) \\ &= \frac{1}{23.21} (6.5 + 0.003695) \\ &= \frac{6.5037}{23.21} = 0.2803 \end{aligned}$$

$$\frac{\Theta_m(s)}{E_a(s)} = \frac{0.0005601}{s(s + 0.2803)}$$

Final Transfer Function:

$$\boxed{\frac{\Theta_m(s)}{E_a(s)} = \frac{0.0005601}{s^2 + 0.2803s}}$$

GEAR SYSTEM



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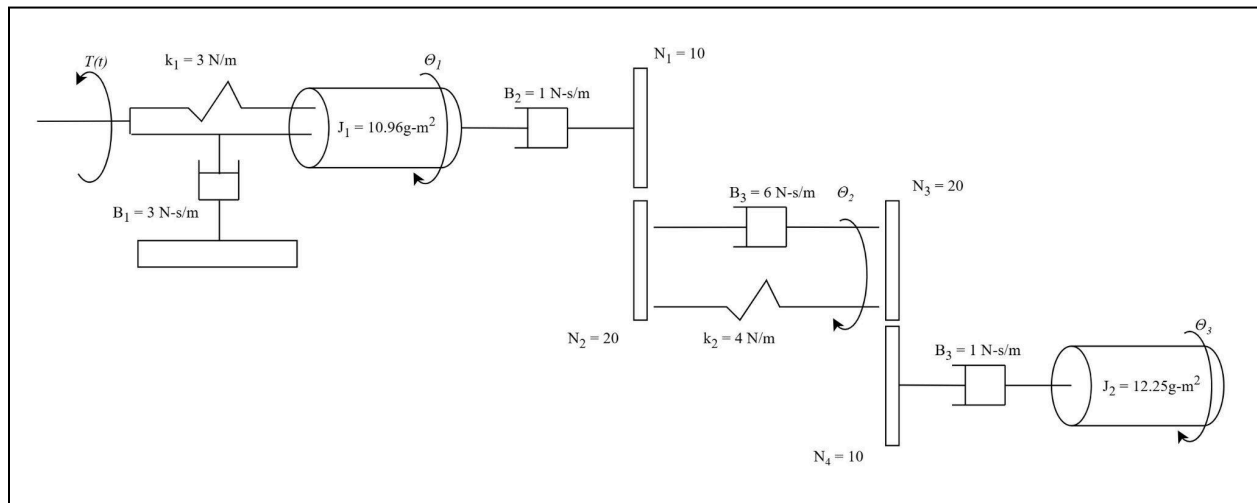
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The gear system model of this project mainly represents how the pulleys and the timing belt gear with each other. The driver pulley J1 has gear movement with the timing belt. Then the timing belt also presents a gear movement with the driven pulley J2.



“figure 4.5. Gear System”

$$I_{eq} = J_1 + \left(\frac{N_1}{N_2}\right)^2 \left[J_{mid} + \left(\frac{N_4}{N_3}\right)^2 J_2 \right]$$

$$I_{eq} = J_1 + \left(\frac{N_1}{N_2}\right)^2 \left(\frac{N_4}{N_3}\right)^2 J_2$$

$$\frac{N_1}{N_2} = \frac{10}{20} = \frac{1}{2}, \frac{N_4}{N_3} = \frac{10}{20} = \frac{1}{2}$$



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$$J_{eq} = J_1 + \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 J_2$$

$$= 10.96 + \frac{1}{16} (12.25)$$

$$= 10.96 + 0.766$$

$$J_{eq} = 11.726$$

$$B_{eq,2} = \left(\frac{1}{2}\right)^2 B_2$$

$$= \frac{1}{4} (1)$$

$$= 0.25$$

$$B_{total} = B_1 + B_{eq,2} + \left(\frac{N_1}{N_2}\right)^2 B_3 + \left(\frac{N_1 N_4}{N_2 N_3}\right)^2 B_4$$

$$= 3 + 0.25 + 1.5 + 0.0625$$



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$$B_{total} = 4.8125$$

$$k_{eq} = \left(\frac{N_1}{N_2}\right)^2 k_2$$

$$= \frac{1}{4}(4)$$

$$k_{eq} = 1$$

$$k_{total} = k_1 + k_{eq}$$

$$= 3 + 1$$

$$k_{total} = 4$$

$$J_{eq}s^2\Theta_1 + B_{eq}s\Theta_1 + k_{eq}\Theta_1 = T(s)$$

$$\Theta_1(11.726s^2 + 4.8125s + 4) = T(s)$$

$$\Theta_3 = \frac{N_1N_4}{N_2N_3}\Theta_1$$



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$$= \frac{1}{4} \Theta_1$$

Final Transfer Function

$$\frac{\Theta_3(S)}{T(S)} = \frac{0.25}{11.726s^2 + 4.8125s + 4}$$

4.2 State-Space Representation

State-Space Representation for the Translational Dynamic System

$$\frac{X_2(s)}{F(s)} = \frac{134.26s^4 + 174.72s^3 + 102.42s^2 + 30s + 4}{s(26042.49s^5 + 35501.62s^4 + 22291.22s^3 + 7333.42s^2 + 1236.72s + 60)}$$

$$D(s) = 26042.49s^5 + 35501.62s^4 + 22291.22s^3 + 7333.42s^2 + 1236.72s + 60$$

$$s^5 + 1.363s^4 + 0.8559s^3 + 0.2816s^2 + 0.04748s + 0.002304$$



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$$N(s) = 134.26s^4 + 174.72s^3 + 102.42s^2 + 30s + 4$$

$$b_4 = 0.005155, b_3 = 0.006709, b_2 = 0.003933, b_1 = 0.001152, b_0 = 0.0001536$$

$$\dot{x}_1 = x_2, \dot{x}_2 = x_3, \dot{x}_3 = x_4, \dot{x}_4 = x_5$$

$$\dot{x}_5 = -0.002304x_1 - 0.04748x_2 - 0.2816x_3 - 0.8559x_4 - 1.363x_5 + u$$

State Space Matrices

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ -0.002304 & -0.04748 & -0.2816 & -0.8559 & -1.363 \end{bmatrix}$$
$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, C = [0.0001536 \quad 0.001152 \quad 0.003933 \quad 0.006709 \quad 0.005155], D = [0]$$

$$T(s) = C(sI - A)^{-1}B + D$$

$$sI - A = \begin{bmatrix} s & -1 & 0 & 0 & 0 \\ 0 & s & -1 & 0 & 0 \\ 0 & 0 & s & -1 & 0 \\ 0 & 0 & 0 & s & -1 \\ 0.002304 & 0.04748 & 0.2816 & 0.8559 & s + 1.363 \end{bmatrix}$$



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$$\det(sI - A) = s \cdot M_{11} - (-1) \cdot M_{12}$$

$$M_{11} = \det \begin{bmatrix} s & -1 & 0 & 0 \\ 0 & s & -1 & 0 \\ 0 & 0 & s & -1 \\ 0.04748 & 0.2816 & 0.8559 & s + 1.363 \end{bmatrix}$$

$$= s \cdot \det \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0.2816 & 0.8559 & s + 1.363 \end{bmatrix} - (-1) \cdot \det \begin{bmatrix} 0 & -1 & 0 \\ 0 & s & -1 \\ 0.04748 & 0.8559 & s + 1.363 \end{bmatrix}$$

$$\begin{aligned} & \det \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0.2816 & 0.8559 & s + 1.363 \end{bmatrix} \\ &= s[s(s + 1.363) - (-1)(0.8559)] - (-1)[0(s + 1.363) - (-1)(0.2816)] \\ &= s[s^2 + 1.363s + 0.8559] + 1[0 + 0.2816] \\ &= s^3 + 1.363s^2 + 0.8559s + 0.2816 \end{aligned}$$

$$\det \begin{bmatrix} 0 & -1 & 0 \\ 0 & s & -1 \\ 0.04748 & 0.8559 & s + 1.363 \end{bmatrix}$$

$$\begin{aligned} &= 0 - (-1) \det \begin{bmatrix} 0 & -1 \\ 0.04748 & s + 1.363 \end{bmatrix} + 0 \\ &= +1 \cdot [0(s + 1.363) - (-1)(0.04748)] \\ &= 0.04748 \end{aligned}$$

$$M_{11} = s(s^3 + 1.363s^2 + 0.8559s + 0.2816) + (0.04748)$$

$$M_{11} = s^4 + 1.363s^3 + 0.8559s^2 + 0.2816s + 0.04748$$



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$$M_{12} = \det \begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & s & -1 & 0 \\ 0 & 0 & s & -1 \\ 0.002304 & 0.2816 & 0.8559 & s + 1.363 \end{bmatrix}$$

$$\begin{aligned} &= 0.002304 \cdot (-1)^{4+1} \det \begin{bmatrix} -1 & 0 & 0 \\ s & -1 & 0 \\ 0 & s & -1 \end{bmatrix} \\ &= 0.002304 \cdot (-1) \cdot [(-1)((-1)(-1) - 0) - 0 + 0] \\ &= -0.002304 \cdot [(-1)(1)] \\ &= -0.002304 \cdot (-1) = 0.002304 \end{aligned}$$

$$\begin{aligned} \det(sI - A) &= s \cdot M_{11} + 1 \cdot M_{12} \\ &= s(s^4 + 1.363s^3 + 0.8559s^2 + 0.2816s + 0.04748) + 0.002304 \\ &= s^5 + 1.363s^4 + 0.8559s^3 + 0.2816s^2 + 0.04748s + 0.002304 \end{aligned}$$

$$26042.49s^5 + 35501.62s^4 + 22291.22s^3 + 7333.42s^2 + 1236.72s + 60$$

$$\begin{aligned} N(s) &= c_1 + c_2s + c_3s^2 + c_4s^3 + c_5s^4 \\ &= 0.0001536 + 0.001152s + 0.003933s^2 + 0.006709s^3 + 0.005155s^4 \end{aligned}$$

$$\frac{X_2(s)}{F(s)} = \frac{134.26s^4 + 174.72s^3 + 102.42s^2 + 30s + 4}{s(26042.49s^5 + 35501.62s^4 + 22291.22s^3 + 7333.42s^2 + 1236.72s + 60)}$$



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State-Space Representation for Rotational Dynamic System

$$\frac{\Theta_4(s)}{T(s)} = \frac{18s^2 + 72s + 72}{1377.51s^6 + 5116.67s^5 + 7575.69s^4 + 6300.54s^3 + 3686.40s^2 + 1224s + 288}$$

$$s^6 + 3.714s^5 + 5.499s^4 + 4.574s^3 + 2.677s^2 + 0.8886s + 0.2091$$

$$b_2 = 0.01307, b_1 = 0.05228, b_0 = 0.05228$$

$$\dot{x}_1 = x_2, \dot{x}_2 = x_3, \dot{x}_3 = x_4, \dot{x}_4 = x_5, \dot{x}_5 = x_6$$

$$\dot{x}_6 = -0.2091x_1 - 0.8886x_2 - 2.677x_3 - 4.574x_4 - 5.499x_5 - 3.714x_6 + u$$

State Space Matrices



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$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -0.2091 & -0.8886 & -2.677 & -4.574 & -5.499 & -3.714 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, C = [0.05228 \quad 0.05228 \quad 0.01307 \quad 0 \quad 0 \quad 0], D = [0]$$

$$T(s) = C(sI - A)^{-1}B + D$$

$$sI - A = \begin{bmatrix} s & -1 & 0 & 0 & 0 & 0 \\ 0 & s & -1 & 0 & 0 & 0 \\ 0 & 0 & s & -1 & 0 & 0 \\ 0 & 0 & 0 & s & -1 & 0 \\ 0 & 0 & 0 & 0 & s & -1 \\ 0.2091 & 0.8886 & 2.677 & 4.574 & 5.499 & s + 3.714 \end{bmatrix}$$

$$\det(sI - A) = s \cdot M_{11} + 1 \cdot M_{12}$$

$$M_{11} = \det \begin{bmatrix} s & -1 & 0 & 0 & 0 \\ 0 & s & -1 & 0 & 0 \\ 0 & 0 & s & -1 & 0 \\ 0 & 0 & 0 & s & -1 \\ 0.8886 & 2.677 & 4.574 & 5.499 & s + 3.714 \end{bmatrix}$$

$$= s \cdot \det \begin{bmatrix} s & -1 & 0 & 0 \\ 0 & s & -1 & 0 \\ 0 & 0 & s & -1 \\ 2.677 & 4.574 & 5.499 & s + 3.714 \end{bmatrix} + 1 \cdot \det \begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & s & -1 & 0 \\ 0 & 0 & s & -1 \\ 0.8886 & 4.574 & 5.499 & s + 3.714 \end{bmatrix}$$



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$$= s \det \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 4.574 & 5.499 & s + 3.714 \end{bmatrix} + 1 \det \begin{bmatrix} 0 & -1 & 0 \\ 0 & s & -1 \\ 2.677 & 5.499 & s + 3.714 \end{bmatrix}$$

$$= s[s(s + 3.714) - (-1)(5.499)] + 1[0(s + 3.714) - (-1)(4.574)]$$

$$= s[s^2 + 3.714s + 5.499] + 4.574$$

$$= s^3 + 3.714s^2 + 5.499s + 4.574$$

$$s(s^3 + 3.714s^2 + 5.499s + 4.574) = s^4 + 3.714s^3 + 5.499s^2 + 4.574s$$

$$= 2.677 \cdot (-1)^{3+1} \det \begin{bmatrix} -1 & 0 \\ s & -1 \end{bmatrix} = 2.677(1) = 2.677$$

$$s^4 + 3.714s^3 + 5.499s^2 + 4.574s + 2.677$$

$$= 0.8886 \cdot (-1)^{4+1} \det \begin{bmatrix} -1 & 0 & 0 \\ s & -1 & 0 \\ 4.574 & 5.499 & s + 3.714 \end{bmatrix}$$

$$= -0.8886 \cdot [(-1)((-1)(s + 3.714) - 0) - 0 + 0]$$

$$= -0.8886 \cdot (-1)(-(s + 3.714))$$

$$= -0.8886 \cdot (-(s + 3.714)) \cdot (-1)$$

$$= -0.8886(s + 3.714)$$

$$\det \begin{bmatrix} -1 & 0 & 0 \\ s & -1 & 0 \\ 4.574 & 5.499 & s + 3.714 \end{bmatrix} = (-1)[(-1)(s + 3.714) - 0] = (-1)(-(s + 3.714))$$

$$= (s + 3.714)$$



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$$(-0.8886)(s + 3.714) = -0.8886s - 3.300$$

$$M_{11} = s(s^4 + 3.714s^3 + 5.499s^2 + 4.574s + 2.677) + (-0.8886s - 3.300)$$

$$M_{11} = s^5 + 3.714s^4 + 5.499s^3 + 4.574s^2 + 2.677s - 0.8886s - 3.300$$

$$M_{11} = s^5 + 3.714s^4 + 5.499s^3 + 4.574s^2 + 1.788s - 3.300$$

$$\det(sI - A) = s^6 + 3.714s^5 + 5.499s^4 + 4.574s^3 + 2.677s^2 + 0.8886s + 0.2091$$

$$1377.51s^6 + 5116.67s^5 + 7575.69s^4 + 6300.54s^3 + 3686.40s^2 + 1224s + 288$$

$$N(s) = c_1 + c_2s + c_3s^2 = 0.05228 + 0.05228s + 0.01307s^2$$

$$\frac{\Theta_4(s)}{T(s)} = \frac{18s^2 + 72s + 72}{1377.51s^6 + 5116.67s^5 + 7575.69s^4 + 6300.54s^3 + 3686.40s^2 + 1224s + 288}$$

State-Space Representation for the Electromechanical Dynamic System

$$\frac{\Theta_m(s)}{E_a(s)} = \frac{0.0005601}{s^2 + 0.2803s} = \frac{0.0005601}{s(s + 0.2803)}$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -0 \cdot x_1 - 0.2803x_2 + u$$

State Space Matrices

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -0.2803 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = [0.0005601 \quad 0], D = [0]$$



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$$T(s) = C(sI - A)^{-1}B + D$$

$$sI - A = \begin{bmatrix} s & -1 \\ 0 & s + 0.2803 \end{bmatrix}$$

$$\det(sI - A) = s(s + 0.2803) = s^2 + 0.2803s$$

$$\text{adj}(sI - A) = \begin{bmatrix} s + 0.2803 & 1 \\ 0 & s \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s^2 + 0.2803s} \begin{bmatrix} s + 0.2803 & 1 \\ 0 & s \end{bmatrix}$$

$$C(sI - A)^{-1}B = \frac{1}{s^2 + 0.2803s} [0.0005601 \quad 0] \begin{bmatrix} s + 0.2803 & 1 \\ 0 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\frac{1}{s^2 + 0.2803s} [0.0005601(s + 0.2803) \quad 0.0005601] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{0.0005601}{s^2 + 0.2803s}$$

$$\boxed{\frac{\Theta_m(s)}{E_a(s)} = \frac{0.0005601}{s^2 + 0.2803s}}$$

State-Space Representation for the Gear Dynamic System

$$\frac{\Theta_3(s)}{T(s)} = \frac{0.25}{11.726s^2 + 4.8125s + 4}$$

$$\frac{0.02132}{s^2 + 0.4104s + 0.3411}$$



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$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -0.3411x_1 - 0.4104x_2 + u$$

State Space Matrices

$$A = \begin{bmatrix} 0 & 1 \\ -0.3411 & -0.4104 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = [0.02132 \quad 0], D = [0]$$

$$T(s) = C(sI - A)^{-1}B + D$$

$$sI - A = \begin{bmatrix} s & -1 \\ 0.3411 & s + 0.4104 \end{bmatrix}$$

$$\det(sI - A) = s(s + 0.4104) + 0.3411 = s^2 + 0.4104s + 0.3411$$

$$\text{adj}(sI - A) = \begin{bmatrix} s + 0.4104 & 1 \\ -0.3411 & s \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{s^2 + 0.4104s + 0.3411} \begin{bmatrix} s + 0.4104 & 1 \\ -0.3411 & s \end{bmatrix}$$

$$C(sI - A)^{-1}B = \frac{[0.02132 \quad 0] \begin{bmatrix} s + 0.4104 & 1 \\ -0.3411 & s \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}}{s^2 + 0.4104s + 0.3411}$$

$$= \frac{[0.02132(s + 0.4104) \quad 0.02132] \begin{bmatrix} 0 \\ 1 \end{bmatrix}}{s^2 + 0.4104s + 0.3411} = \frac{0.02132}{s^2 + 0.4104s + 0.3411}$$

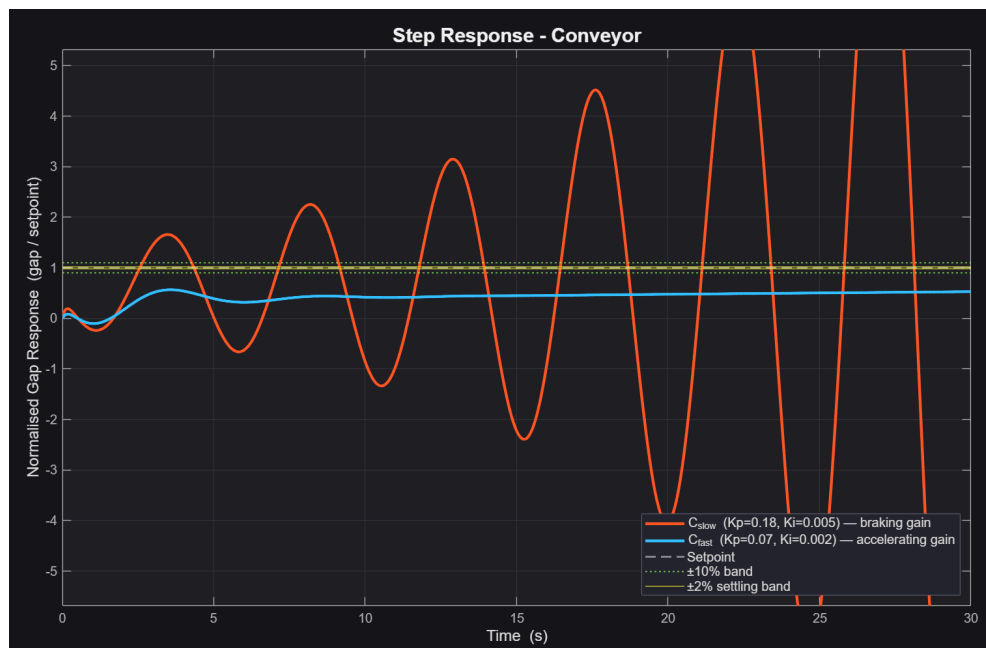
$$\boxed{\frac{\Theta_3(s)}{T(s)} = \frac{0.25}{11.726s^2 + 4.8125s + 4}}$$



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V. Time Response Analysis

The zoomed-out step response graph illustrates the overall behavior of the conveyor belt PI control system over a longer period of operation. The response shows how the controller continuously adjusts the conveyor speed in reaction to spacing errors detected by the IR proximity sensors. Oscillatory behavior can be observed due to repeated corrections made by the PI controller as the system attempts to maintain the desired item spacing set point. The increasing oscillation magnitude demonstrates the sensitivity of the system to sustained disturbances and varying item intervals along the conveyor belt.



“figure 5.1. Step Response zoomed out”



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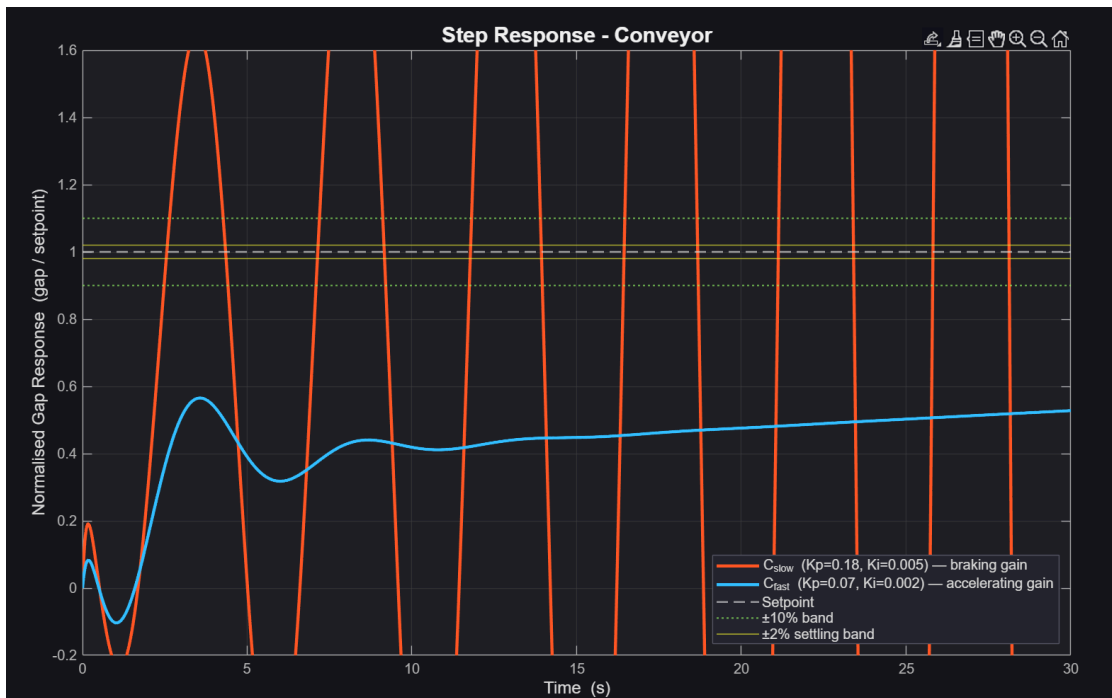
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The zoomed-in step response provides a closer observation of the transient response characteristics of the system immediately after a spacing disturbance occurs. The graph shows the rise time, overshoot, and settling behavior of the PI controller as it attempts to stabilize the conveyor discharge interval. Initially, the system reacts aggressively to spacing errors, causing temporary oscillations before gradually approaching the desired operating condition. This demonstrates the controller's capability to dynamically regulate belt speed in response to changing object spacing conditions.



“figure 5.2. Step Response zoomed in”



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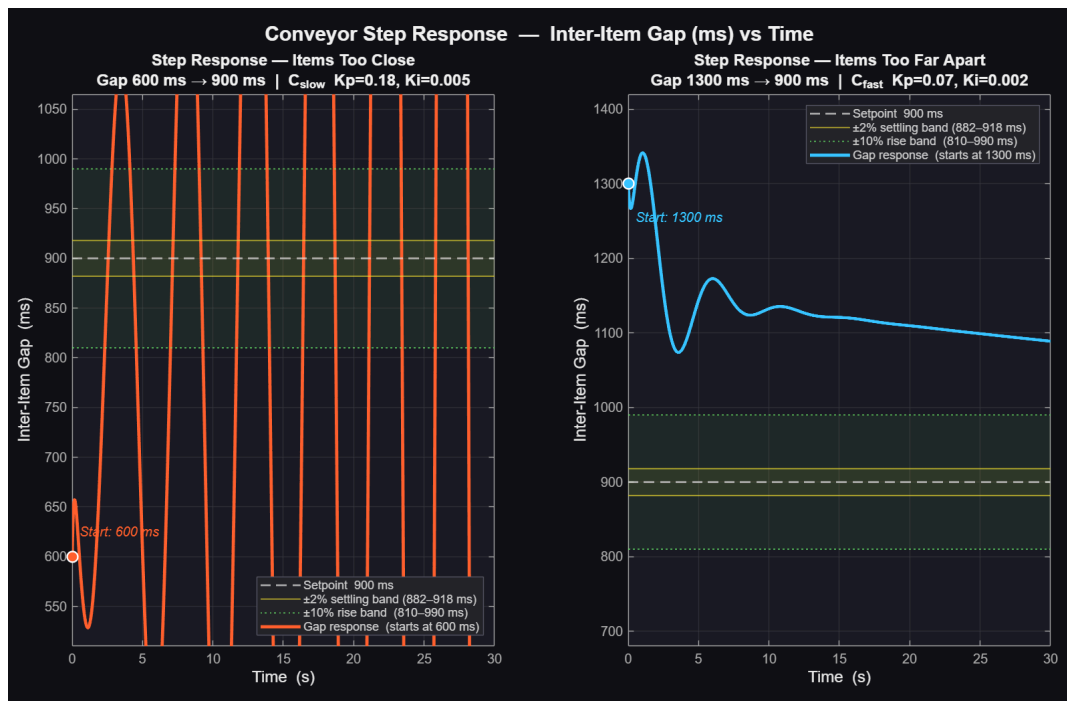
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This figure compares the system response under two operating conditions: items positioned too close together and items positioned farther apart than the target spacing. When objects are too close, the controller reduces conveyor speed to prevent congestion at the discharge area. Conversely, when objects are too far apart, the controller increases belt speed to compensate for delayed delivery. The graph demonstrates the adaptability of the PI controller in maintaining consistent inventory discharge despite varying item intervals.



“figure 5.3. Step Response (close & far apart item scenarios)”



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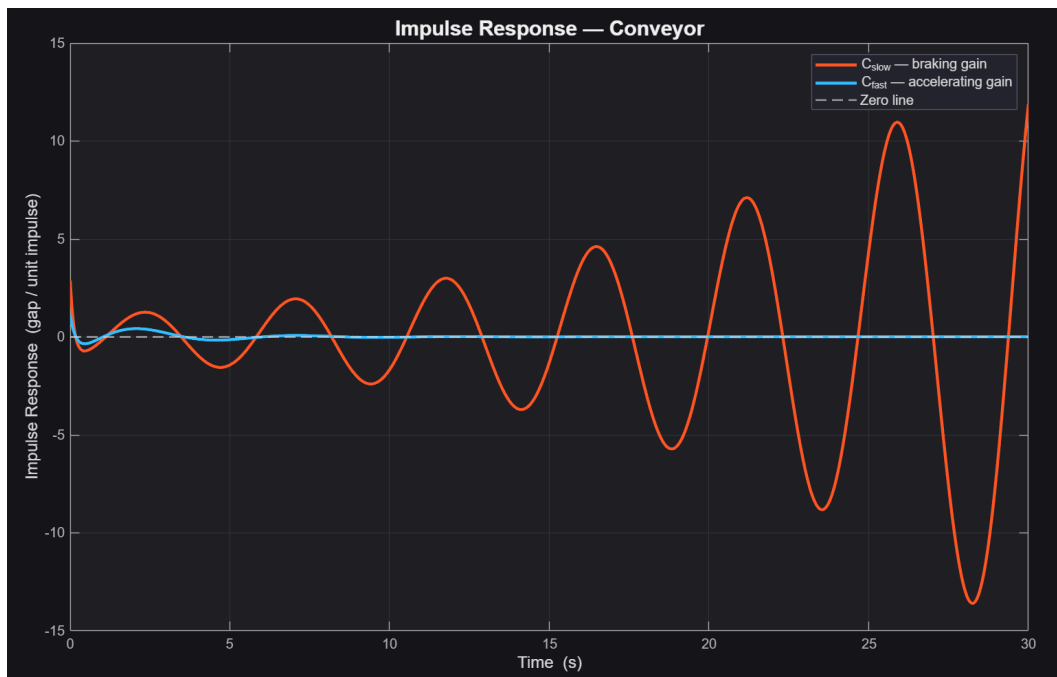
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The impulse response graph represents the reaction of the conveyor system to a sudden spacing disturbance or abrupt detection event. The response exhibits oscillatory characteristics caused by the controller attempting to quickly correct the error introduced into the system. Over time, the controller works to reduce the effect of the disturbance and stabilize the conveyor operation. This behavior highlights the responsiveness of the closed-loop PI control system in handling sudden changes in item spacing conditions.



“figure 5.4. Impulse Response”



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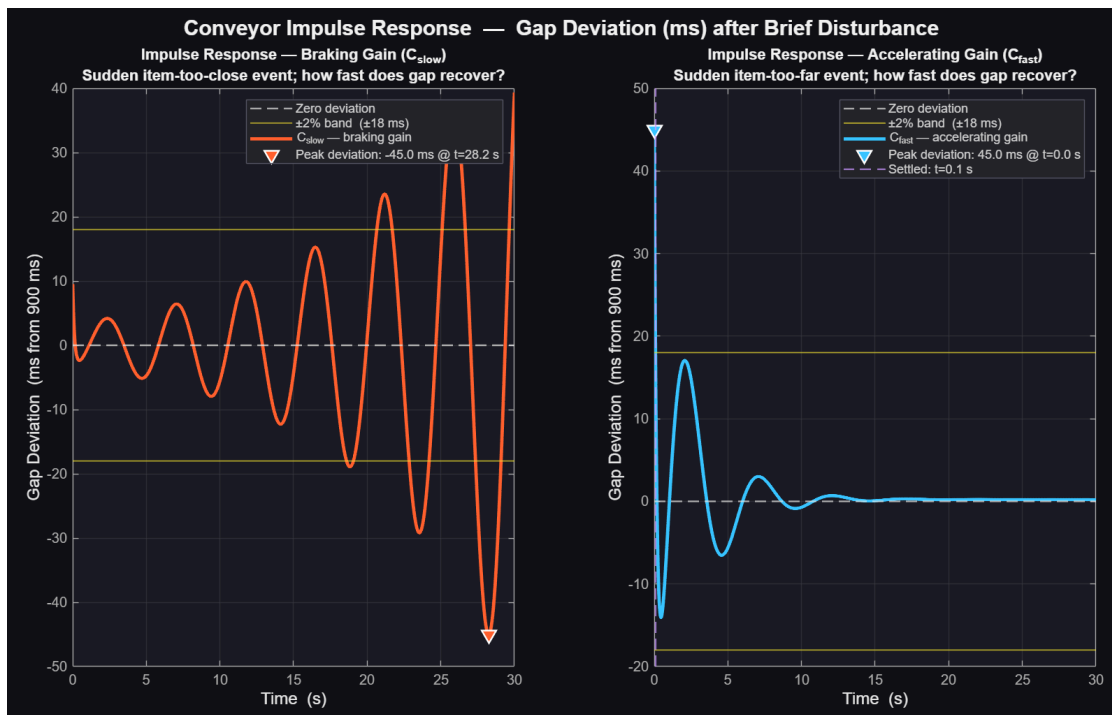
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This graph illustrates the impulse response of the conveyor system for both close-spacing and far-spacing disturbances. The system produces different corrective actions depending on the detected spacing condition. Larger corrective responses are observed when items are significantly closer or farther from the desired set point, indicating stronger controller action to restore proper spacing. The graph confirms that the controller can react dynamically to sudden disturbances while maintaining operational stability.



“figure 5.5. Impulse Response (close & far item scenarios)”



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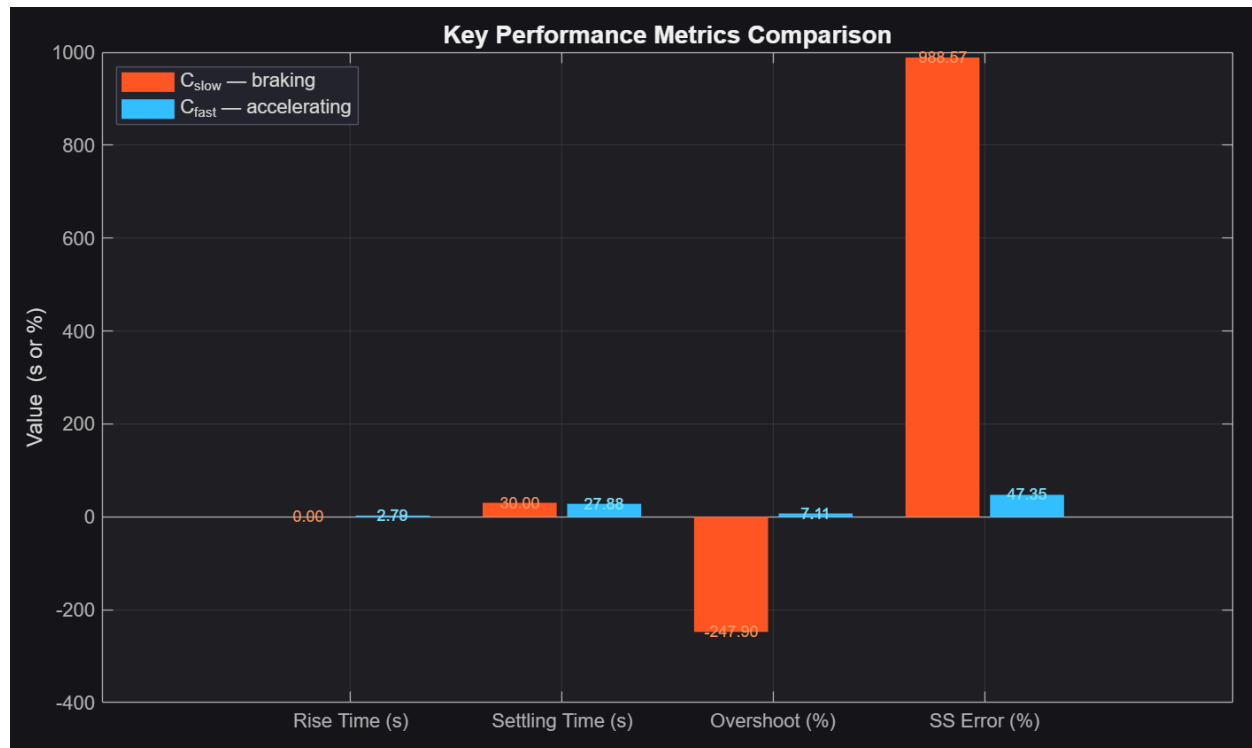
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The key performance metrics comparison summarizes important transient response characteristics of the conveyor control system, including rise time, settling time, overshoot, and steady-state error. The comparison highlights the differences in controller behavior between close-item and far-item operating conditions. Variations in these metrics indicate how the PI controller responds differently depending on the magnitude and direction of the spacing error. Overall, the graph demonstrates the effectiveness of the controller in minimizing steady-state error while maintaining acceptable response performance.



“figure 5.6. Key Performance Metrics Comparison”



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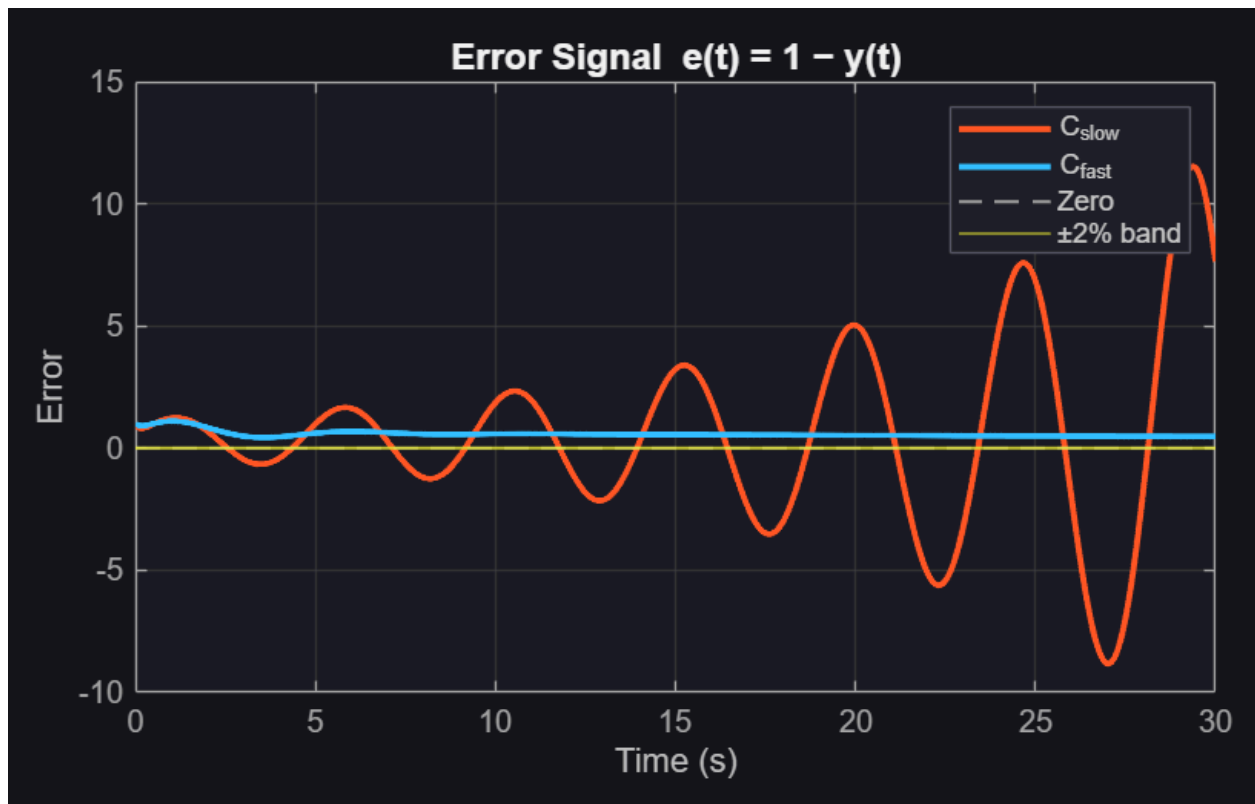
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The error signal graph shows the difference between the desired spacing set point and the actual measured spacing over time. Positive and negative error values indicate whether items are too close together or too far apart, respectively. The oscillating nature of the error signal reflects the continuous corrective action performed by the PI controller as it regulates conveyor speed. Although transient oscillations are present, the controller continuously works to minimize the error and maintain consistent item discharge spacing.



“figure 5.7. Error Signal”



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VI. MATLAB Implementation

Modeling using MATLAB/Simulink

The conveyor belt PI control system was modeled and simulated using MATLAB/Simulink to validate the mathematical model and evaluate controller performance prior to physical prototype implementation. The Simulink environment provided a platform for implementing the closed-loop control architecture derived from the system's electromechanical transfer function, allowing observation of system behavior under various spacing conditions without the constraints of physical hardware limitations.

The simulation model was built around the derived electromechanical transfer function:

$$\frac{\Theta_m(s)}{E_a(s)} = \frac{0.0005601}{s^2 + 0.2803s} = \frac{0.0005601}{s(s + 0.2803)}$$

This transfer function served as the plant model within the Simulink block diagram, representing the dynamic relationship between the applied armature voltage and the resulting motor shaft angular displacement. The transfer function captures the dominant dynamic behavior of the DC gear motor driving the conveyor belt, including the effects of combined inertia and damping derived from the gear system analysis.

Figure 6.1 presents the complete Simulink block diagram of the closed-loop conveyor PI control system. The simulation architecture consists of several interconnected subsystems that mirror the physical hardware configuration of the prototype. A reference setpoint of 900 ms representing the target inter-item gap is fed into a summing junction where the error signal is



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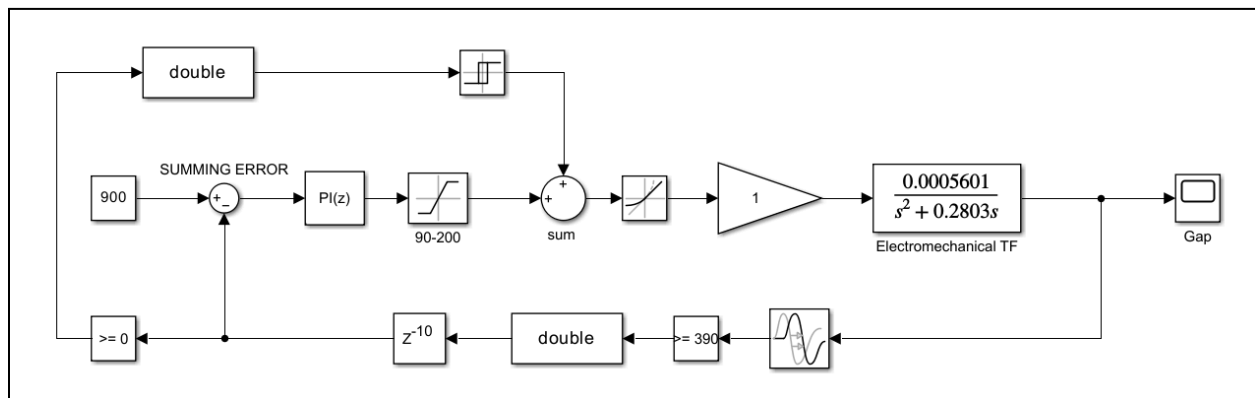
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computed by subtracting the measured gap feedback. The error signal is then processed by a discrete-time PI controller block, labeled $PI(z)$, which computes the corrective control output based on the asymmetric gain structure of the implemented controller. The PI controller output is subsequently passed through a saturation block that enforces the defined PWM operating limits of 90 to 200, preventing actuator saturation and protecting the motor from out-of-range commands. A unity gain summing block combines the PI output before it is applied to the electromechanical transfer function block representing the conveyor motor plant. The output of the plant block corresponds to the inter-item gap measured at the discharge sensor IR2. A transport delay element represented by the Z^{-10} block was incorporated into the feedback path to simulate the belt's inherent transport delay between the entry sensor IR1 and the discharge sensor IR2, which is approximately two seconds under nominal operating conditions. A threshold comparator block checking whether the gap is greater than or equal to zero was also included to implement the conditional logic governing the asymmetric gain selection between braking and accelerating controller modes. The feedback signal is then routed back to the summing junction to close the control loop, and a scope block labeled "Gap" captures the inter-item gap response over time for analysis.



“figure 6.1. simulink simulation diagram”



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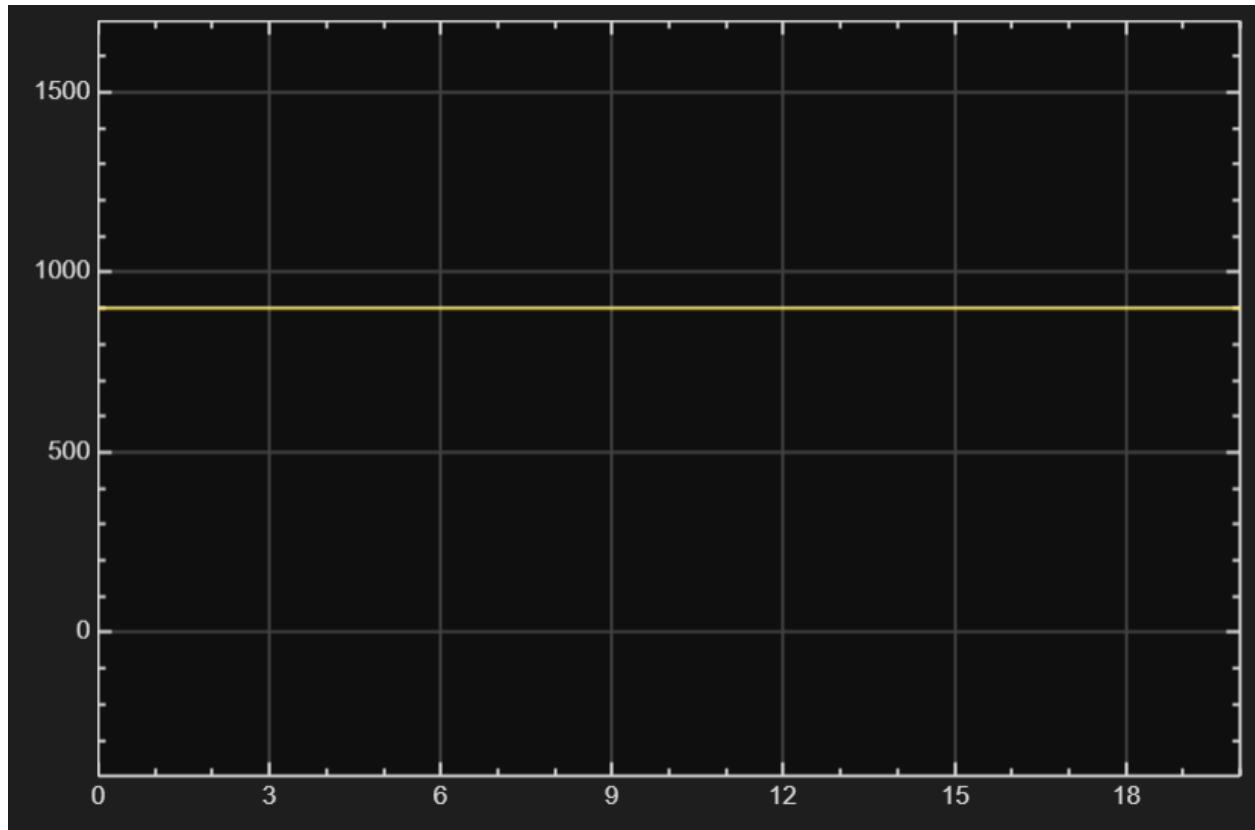
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Figure 6.2 shows the Simulink simulation output under ideal steady-state conditions where the inter-item gap converges to approximately 900 ms and remains stable throughout the simulation duration of approximately 18 seconds. The output trace exhibits a flat horizontal response near the setpoint value, confirming that under nominal conditions with items introduced at the target interval, the PI controller maintains the conveyor at its default PWM output of 110 with negligible corrective action required. This result validates that the model correctly implements steady-state behavior and that the controller does not introduce unnecessary oscillations when the system is operating at its desired operating point.



“figure 6.2. simulink diagram output”



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MATLAB Digital Twin

Figure 6.3 presents the MATLAB Digital Twin interface developed to complement the Simulink model and serve as an interactive simulation environment that mirrors the physical conveyor prototype. The digital twin, designated as version 5, replicates the full system configuration including the conveyor belt visualization, dual IR sensor positions, real-time LCD display output, live PI signal plots, and an operator control panel. The interface displays real-time values for PWM output, inter-item gap in milliseconds, item count, and PI error, mirroring the data visible on the physical prototype's I2C LCD display. The control panel provides START, STOP, and RESET functions, as well as a manual mode toggle and an item injection button for simulating item entry at IR1. Adjustable slider controls for item spacing interval, Kp slow, Ki slow, Kp fast, and Ki fast allow the operator to tune the asymmetric PI gains interactively and observe the resulting system response in real time through the three live signal plots showing inter-item gap, PWM output, and PI error over a rolling 30-second window.



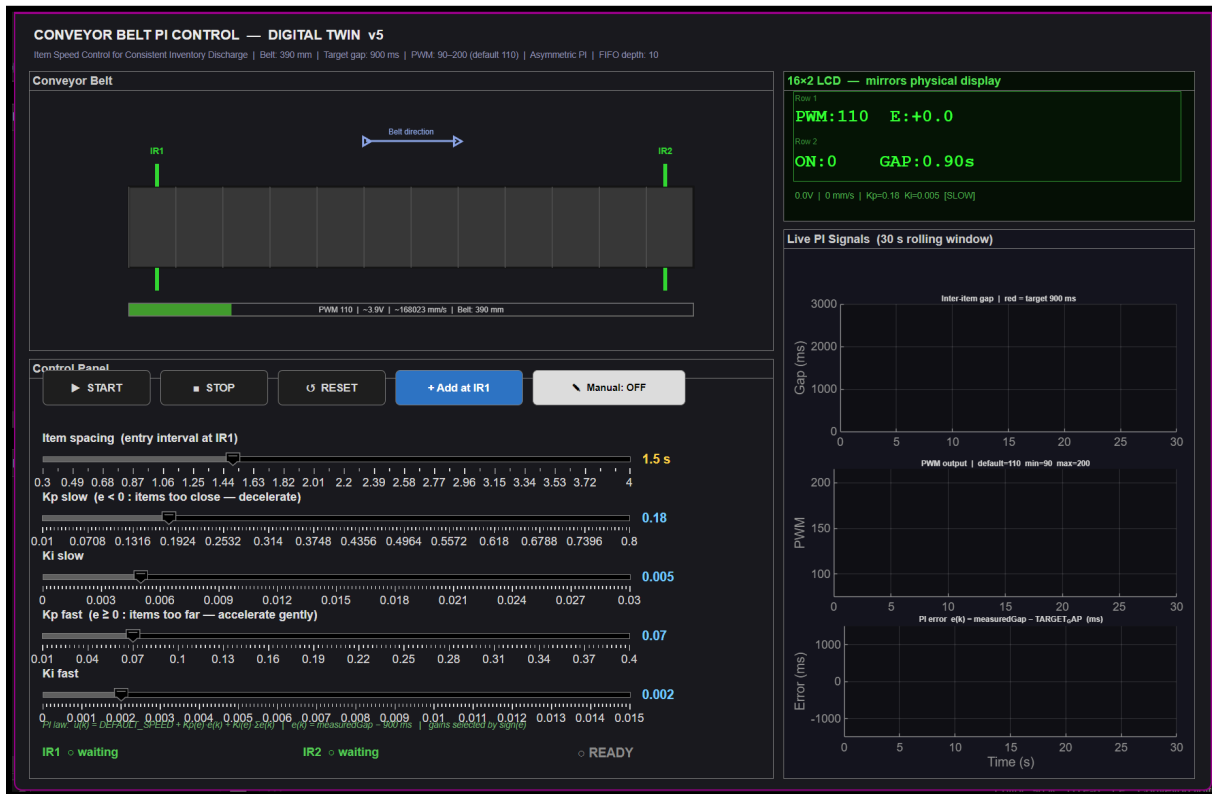
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“figure 6.3. MatLab digital twin”

Figure 6.4 illustrates the digital twin operating in automatic mode under an ideal spacing condition where two items are placed on the belt at approximately the target gap interval of 900 ms. The LCD display panel on the right side of the interface shows a PWM value of 119 with an error of +1.0, an item count of 2, and a measured gap of 1.00 seconds. The live signal plots confirm stable gap tracking near the setpoint with minimal PWM variation and a near-zero error signal, demonstrating that the controller correctly maintains steady-state operation when items are introduced at the desired interval.



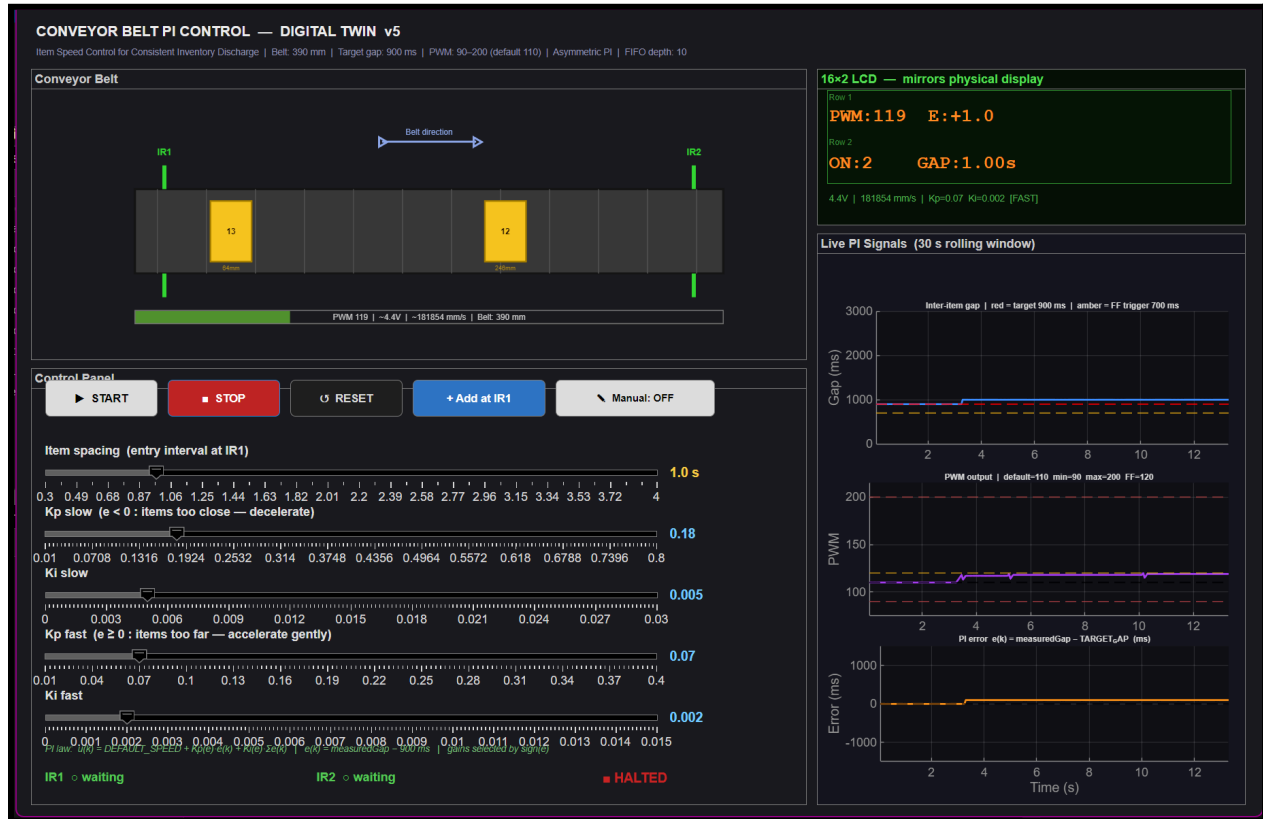
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“figure 6.4. Matlab Digital twin auto mode (ideal state)”

Figure 6.5 shows the digital twin under the close-spacing scenario in manual mode, where two items are visible on the conveyor belt positioned closer together than the target threshold. This condition corresponds to a measured gap below 900 ms, triggering the braking gain set of $K_p = 0.18$ and $K_i = 0.005$. The conveyor belt visualization shows the items in close proximity, and the system is prepared to apply a speed reduction corrective response upon detection at IR2.



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The accompanying graph data for the close scenario, shown in the figure labeled as the close scenario graph data, presents the time history of the inter-item gap, PWM output, and PI error over approximately six seconds. The gap plot shows the measured gap dipping below the 900 ms setpoint reference line before the PI controller reduces the PWM output, which is reflected in the PWM trace dropping below the default value of 110. The error plot confirms a corresponding negative error during the gap undershoot period, followed by recovery toward zero as the corrective action takes effect. The feed-forward trigger at 700 ms, represented by the amber reference line in the gap plot, is also visible, demonstrating the pre-emptive speed reduction mechanism activated when the entry gap is detected below the feed-forward threshold.

The far-spacing scenario shown in the subsequent digital twin figure presents the opposite condition, where two items are placed with a gap significantly exceeding the target interval. In this case the accelerating gain set of $K_p = 0.07$ and $K_i = 0.002$ is activated, and the PWM output increases above the default value to reduce the excessive discharge interval. The corresponding graph data for the far scenario shows the inter-item gap rising above the setpoint reference during the period of large spacing, with the PWM trace increasing above 110 in response. The error signal displays a positive deviation during this period before returning toward zero as the controller brings the gap back toward the 900 ms target. The settled response visible in the error plot confirms the integral action of the PI controller successfully eliminates steady-state offset under the far-spacing condition.



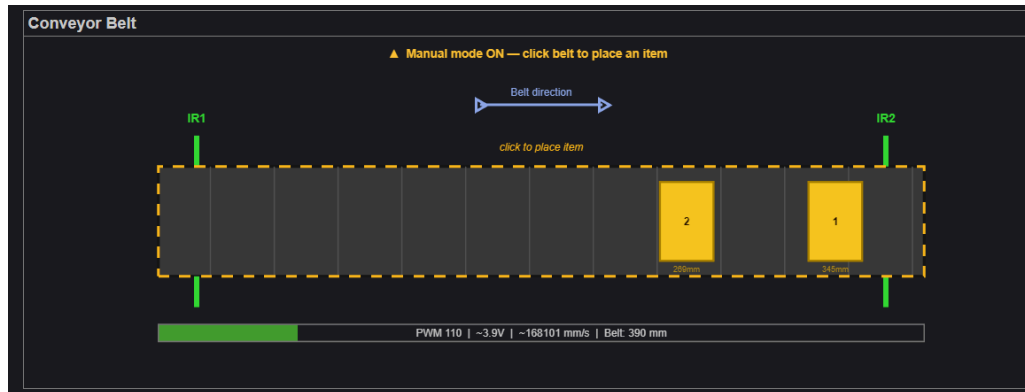
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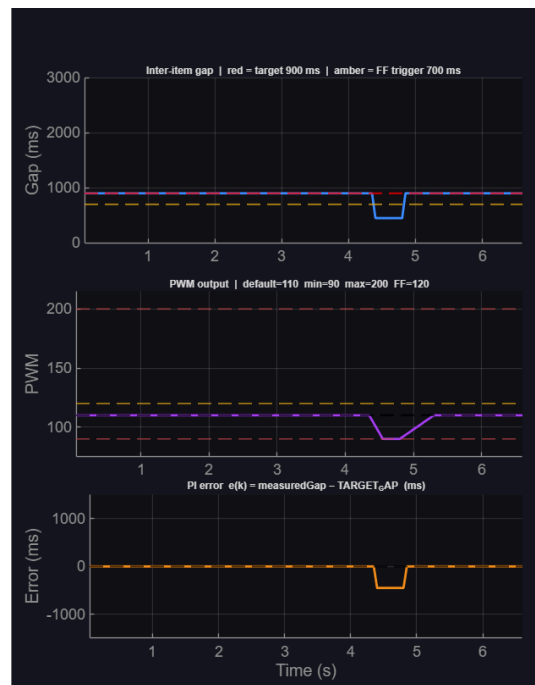
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“figure 6.5. Matlab Digital twin (close scenario)”



“figure 6.4. Matlab Digital twin close scenario graph data”



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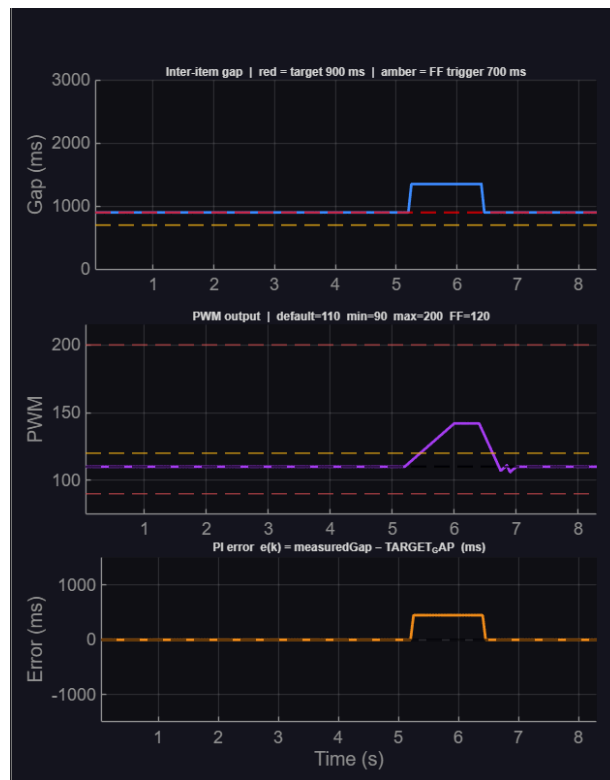
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“figure 6.4. Matlab Digital twin far scenario”



“figure 6.4. Matlab Digital twin far scenario graph data”

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Controller Tuning

The PI controller implemented in this system employs an asymmetric gain structure rather than a conventional fixed-gain approach, reflecting the physically different dynamics encountered during deceleration and acceleration of the conveyor belt. Controller tuning was performed iteratively using the MATLAB Digital Twin interface, which allowed real-time observation of gap response, PWM output, and error signal behavior while adjusting gain values through the interactive sliders prior to flashing the parameters onto the Arduino firmware.

Two distinct gain sets were defined based on the sign of the error signal. When the measured gap falls below the 900 ms target, indicating items are too close together, the braking gain set of $K_p = 0.18$ and $K_i = 0.005$ is applied. The higher proportional gain in this configuration produces a stronger and faster speed reduction response, reflecting the greater urgency of preventing item congestion at the discharge region. When the measured gap exceeds the target, indicating items are too far apart, the accelerating gain set of $K_p = 0.07$ and $K_i = 0.002$ is engaged. The lower gain values in this configuration produce a gentler speed increase, avoiding overshoot during acceleration and preventing the belt from running excessively fast between corrections.

The default PWM output was set to 110, corresponding to an applied voltage of approximately 3.9 V to the DC gear motor and a nominal belt speed of approximately 168 mm per second under no-load conditions. The PWM output was clamped within the range of 90 to 200 to protect the motor from operating below its minimum effective speed threshold or exceeding its rated capacity. The integral term was bounded by a hard clamp of ± 1000 , limiting the maximum integral contribution to ± 5 PWM units and preventing integral windup during sustained error conditions. Anti-windup was implemented through conditional integration, where



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the integral accumulator is only updated when the controller output is not simultaneously saturated in the same direction as the error.

An asymmetric PWM ramp rate was also incorporated into the actuator model, where the conveyor decelerates at a rate of 5 PWM units per step while it accelerates at only 2 PWM units per step. This 2.5:1 deceleration-to-acceleration ratio prevents sudden item pile-up that could occur if the belt slowed too abruptly during a braking correction, ensuring smooth and controlled speed transitions throughout normal operation.

A feed-forward trigger mechanism was additionally implemented to address the approximately two-second transport delay between IR1 and IR2. When the entry gap detected at IR1 is less than 700 ms, a one-shot pre-emptive speed reduction to a PWM value of 120 is scheduled 1000 ms after the detection event, firing midway through the transport delay before the item reaches IR2. This Smith Predictor-inspired approach allows the system to begin corrective action earlier than the feedback loop alone would permit, reducing the effective impact of the transport delay on system response and improving overall discharge consistency.

VII. Experimental / Prototype Testing

The prototype features a custom-designed 3D-printed conveyor system actuated by a TT DC gear motor. The motor drives the conveyor belt at an adjustable speed regulated by an L298N motor driver module, which accepts PWM signals to vary the output voltage supplied to the motor.

The system logic is programmed and executed on an Arduino Uno microcontroller, which interfaces with two infrared proximity sensors to implement closed-loop speed control. The first sensor, IR1, is positioned at the entry point of the conveyor and is responsible for detecting the



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arrival of incoming objects and recording the inter-item gap interval at the point of entry. The second sensor, IR2, is positioned near the discharge end of the belt and serves as the feedback point for the control system, triggering speed adjustments upon detecting each exiting item to maintain a consistent and uniform discharge rate.

To ensure that each speed correction applied at IR2 corresponds precisely to the gap measurement of the correct item, a First-In First-Out (FIFO) queue structure was implemented in firmware. Gap intervals recorded at IR1 are stored sequentially in the queue and retrieved in the same order as items arrive at IR2, preserving measurement-to-item correspondence even when multiple objects are in transit on the belt simultaneously. The queue supports a capacity of up to ten concurrent items.

Several protective measures are incorporated to ensure safe and reliable operation. Motor speed is bounded within a defined operational range, with a minimum PWM value of 90, a maximum of 200, and a default operating value of 110, preventing the motor from being driven below its minimum effective threshold or beyond its rated capacity. These constraints extend motor service life and reduce the risk of mechanical overload during transient control corrections.

To evaluate the performance of the prototype, three identical test objects in the form of cubes measuring 1.5 cm per side were used as the detection targets throughout all testing procedures. The cubes were introduced onto the entry ramp sequentially and detected by IR1 under four distinct test conditions, each designed to assess a specific aspect of the system's closed-loop speed control response.

The first test established a baseline by introducing the cubes at the target gap interval of 900 ms, representing the ideal spacing condition under which the system is expected to operate with minimal correction. The second test introduced the cubes at an interval below 900 ms,

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simulating an overcrowded entry condition in which items are spaced too closely, requiring the system to reduce belt speed in order to restore the target discharge gap. The third test introduced the cubes at an interval exceeding 900 ms, representing an underpopulated entry condition in which items are spaced too far apart, requiring the system to increase belt speed to recover the target gap. The fourth and final test combined both conditions simultaneously using all three cubes, where one pair of cubes was introduced with a gap below the target threshold and the other pair with a gap above it, evaluating the system's ability to handle mixed spacing conditions within a single batch and correctly apply independent corrections to each item through the FIFO queue.

VIII. Results and Discussion

Analysis of System Performance

The developed prototype successfully demonstrated the implementation of closed-loop PI-based speed control for maintaining consistent object discharge spacing in a conveyor system. By integrating two infrared proximity sensors, an Arduino Uno microcontroller, and a FIFO queue structure, the system was able to monitor object intervals and apply corresponding speed corrections in real time. The prototype utilized a custom-designed 3D-printed conveyor actuated by a TT DC gear motor controlled through an L298N motor driver module. The conveyor speed was dynamically adjusted through PWM control signals generated by the Arduino Uno based on spacing feedback obtained from the infrared sensors. IR1 measured the incoming inter-item gap intervals, while IR2 acted as the feedback sensor responsible for triggering corrective actions near the discharge region of the conveyor. Experimental evaluation was conducted using three identical cubic test objects measuring 1.5 cm on each side. The system was tested under different



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spacing conditions centered around the target discharge interval of 900 ms. The obtained results demonstrated that the controller could effectively compensate for both overcrowded and underpopulated conveyor conditions.

Test scenario	Initial Gap Condition	Controller Response	Conveyor Speed action	Observed System Behaviour
Small gap trial	Gap < 900 ms	Positive spacing correction	Conveyor speed reduced	Discharge spacing gradually increased toward target interval
Large gap trial	Gap > 900 ms	Negative spacing correction	Conveyor speed increased	Discharge spacing gradually decreased toward target interval

During the small-gap trial, the objects were introduced with intervals below the desired 900 ms set point, simulating an overcrowded conveyor condition. Once the reduced spacing was detected, the PI controller responded by decreasing the PWM output supplied to the DC gear motor. This reduced conveyor speed and increased the discharge interval between objects, helping prevent object congestion at the output region. The recorded step and impulse response graphs showed transient oscillations immediately after correction due to the proportional and integral actions of the controller. However, the system gradually stabilized and approached the target operating condition without instability.



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In the large-gap trial, the objects were introduced with intervals exceeding the target spacing, representing delayed item flow conditions. Upon detection at IR2, the controller increased the PWM duty cycle to accelerate the conveyor belt. This reduced the excessive discharge delay between successive objects and allowed the measured spacing to move closer toward the target interval. The system exhibited faster transient response characteristics during acceleration due to the motor's ability to increase speed more rapidly than decelerate under load conditions. Minor overshoot was observed during correction, but the system eventually settled toward steady-state operation. The implemented FIFO queue structure played a significant role in maintaining system reliability during testing. Since multiple objects could simultaneously occupy the conveyor, the queue ensured that each spacing measurement obtained at IR1 was correctly associated with its corresponding object upon arrival at IR2. This prevented incorrect speed corrections and maintained accurate closed-loop operation even under mixed spacing conditions.

Overall, the experimental results indicate that the developed prototype was capable of dynamically regulating conveyor discharge spacing using low-cost embedded hardware and infrared sensing technologies. The controller successfully performed bidirectional corrective actions by slowing the conveyor during overcrowded conditions and accelerating it during underpopulated conditions while maintaining stable system behavior.

Sources of Error

Although the prototype demonstrated successful closed-loop control operation, several factors introduced inaccuracies and limitations during experimentation. One significant source of error was the inherent response limitation of the TT DC gear motor. Due to motor inertia and mechanical lag, changes in conveyor speed were not instantaneous after PWM adjustments were



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applied. This resulted in transient overshoot and oscillatory behavior within the time response graphs, particularly during sudden spacing corrections. Sensor-related limitations also contributed to system inaccuracies. The infrared proximity sensors were sensitive to object alignment, ambient lighting conditions, and object reflectivity. Slight variations in cube positioning or sensor detection timing occasionally produced inconsistent gap measurements, affecting the precision of the controller's corrective response.

Mechanical inconsistencies within the 3D-printed conveyor structure introduced additional disturbances. Minor belt slippage, uneven belt tension, and friction between moving components could alter the effective conveyor speed independently of the controller output. These mechanical effects became more noticeable during rapid acceleration or deceleration conditions. Timing inaccuracies associated with the Arduino Uno microcontroller and software execution also contributed to measurement deviations. Since the controller relied on millisecond timing intervals and sequential program execution, small processing delays could accumulate during multi-object operation, especially when multiple queue operations and sensor detections occurred simultaneously. Another limitation was the simplified tuning of the PI controller gains. While the selected gain values produced stable operation, the controller was not fully optimized for all possible operating conditions. As a result, small steady-state errors and temporary oscillations remained present during system response. More advanced tuning methods or the implementation of additional derivative control could potentially improve transient response performance and reduce overshoot.

Despite these limitations, the prototype successfully demonstrated the feasibility of using closed-loop PI control for conveyor spacing regulation. The observed errors were generally minor and did not prevent the system from achieving its primary objective of maintaining a more consistent object discharge interval.



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IX. Conclusion

This project successfully designed, modeled, simulated, and physically implemented a closed-loop PI-based conveyor speed control system for consistent inventory discharge using infrared proximity sensors. Through the integration of mechatronics engineering principles spanning mechanical design, electronics, embedded programming, and control theory, the project demonstrated that a low-cost embedded system is capable of dynamically regulating conveyor belt speed to maintain a consistent inter-item discharge interval under varying input spacing conditions.

Summary of Findings

The mathematical modeling phase established transfer functions and state-space representations for each subsystem of the conveyor system, including the electrical circuit, translational mechanical assembly, rotational mechanical drivetrain, electromechanical motor model, and gear system. The dominant electromechanical transfer function of the form $0.0005601 / (s^2 + 0.2803s)$ was identified as the primary plant model governing motor shaft response to applied armature voltage. State-space matrices derived from the translational and rotational models further characterized the multi-degree-of-freedom dynamic behavior of the conveyor structure, confirming system stability and providing a foundation for controller design.

The asymmetric PI control law implemented in this system used separate gain sets for braking and accelerating conditions, with $K_p = 0.18$ and $K_i = 0.005$ applied during close-spacing corrections and $K_p = 0.07$ and $K_i = 0.002$ applied during far-spacing corrections. This asymmetric structure was motivated by the physically different urgency and dynamic characteristics of deceleration versus acceleration in the conveyor system. Anti-windup conditioning, PWM output clamping between 90 and 200, asymmetric ramp rates favoring faster



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deceleration over acceleration, and a feed-forward pre-compensation trigger were incorporated to improve robustness and reduce the impact of the approximately two-second belt transport delay. A FIFO queue structure implemented in firmware ensured accurate measurement-to-item correspondence when multiple objects simultaneously occupied the conveyor belt.

Time response analysis conducted through MATLAB simulation showed that the accelerating gain configuration produced faster rise time and quicker settling behavior, while the braking configuration exhibited larger overshoot and more pronounced oscillatory transients due to its higher proportional gain. The key performance metrics comparison confirmed that steady-state error under the accelerating condition was significantly lower than under the braking condition, and that both configurations successfully drove the inter-item gap toward the 900 ms target setpoint over time. The MATLAB digital twin validated these behaviors interactively prior to hardware testing and demonstrated reliable correspondence between simulated and physical system responses.

Prototype testing under four experimental conditions — ideal spacing, close spacing, far spacing, and mixed spacing — confirmed that the system performed bidirectional corrective actions correctly in all cases. During the small-gap trial, the controller reduced conveyor speed and gradually increased the discharge interval toward the target. During the large-gap trial, the controller increased belt speed and reduced the excessive discharge delay. The FIFO queue successfully maintained correct correction-to-item pairing during the mixed-spacing trial involving multiple objects with different gap conditions simultaneously on the belt. Despite the presence of minor oscillatory transients, mechanical disturbances, and sensor variability, the system consistently achieved its primary objective of maintaining a more uniform discharge interval compared to uncontrolled operation.



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Effectiveness of Closed-Loop Control

The results of this project affirm that closed-loop control is essential for achieving reliable and adaptive conveyor operation. Unlike open-loop systems that operate at a fixed speed regardless of actual item conditions, the closed-loop architecture employed here continuously monitors real-time spacing feedback and applies corrective action only when deviation from the setpoint is detected. This feedback-driven approach allowed the system to compensate for variable item introduction rates, mechanical inconsistencies, and transient disturbances without requiring manual intervention. The integral action of the PI controller further ensured that residual steady-state errors were minimized over successive correction cycles, improving the overall consistency of discharge spacing across extended operation. The asymmetric gain design and feed-forward compensation demonstrated that purposeful controller customization beyond standard fixed-gain tuning can meaningfully improve performance in systems with asymmetric physical dynamics and significant transport delay.

Industrial Relevance

The principles demonstrated in this project are directly applicable to industrial-scale conveyor automation systems used in manufacturing plants, packaging facilities, warehouses, and distribution centers. In production environments where downstream stations such as robotic arms, packaging machines, labeling equipment, or human operators depend on a predictable and consistent product arrival rate, irregular conveyor discharge can cause congestion, processing delays, equipment overload, and safety risks. The closed-loop speed control approach demonstrated here addresses these challenges by automatically regulating item flow based on real-time sensor feedback, eliminating the need for fixed-speed operation and reducing the likelihood of bottlenecks or gaps in the production stream.



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Furthermore, the low-cost bill of materials totaling approximately PHP 3,946.00, the use of commercially available embedded hardware, and the modular 3D-printed mechanical design demonstrate that effective closed-loop conveyor control need not require expensive industrial programmable logic controllers or proprietary automation hardware. This makes the system concept accessible for small-to-medium scale manufacturing operations and educational demonstrations of industrial automation principles. Future work may explore the integration of a full PID controller with derivative action to further reduce transient overshoot, the use of encoder-based speed feedback to improve motor control precision, and the scaling of the system to handle heavier loads and longer belt configurations representative of real industrial conveyor environments.



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X. Documentation

BILL OF MATERIALS				
Item	Description	Qty	Unit Price (PHP)	Price (PHP)
DC motor	TT Gear Motor	1	49.00	49.00
IR Sensor	HW 201 Infra-Red Proximity sensor	2	49.00	98.00
L298N	Stepper motor drive controller board module	1	99.00	99.00
Switching Power Supply	230V AC to 12V 5A DC converter	1	309.00	309.00
LCD Display with I2c connector	Display to monitor the system real-time	1	149.00	149.00
LM2596	Step-Down Adjustable DC-DC Switching Buck converter	1	55.00	55.00
Arduino Uno R3	Primary Controller	1	449.00	449.00
Sensor Shield v5 for arduino	Provides additional VCC and GND pins	1	99.00	99.00



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1kg 3D Printing Filament PETG	(Polyethylene Terephthalate Glycol) 1.75mm 3D Printing Filament.	1	483.00	483.00
(Small) MTS102 toggle switch	Serves as the toggle switch for system power	1	25.00	25.00
Push Button with Micro Switch	One Red and One Green	2	26.00	52.00
Waterproof plastic case	100x68x50mm Outdoor Waterproof Case Enclosure Plastic Box Electronic Project Junction	1	99.00	99.00
Plywood	300x600x10mm plywood	1	230.00	230.00
Additive manufacturing cost (527.73g)				1,750.00
Total Cost				3,946.00



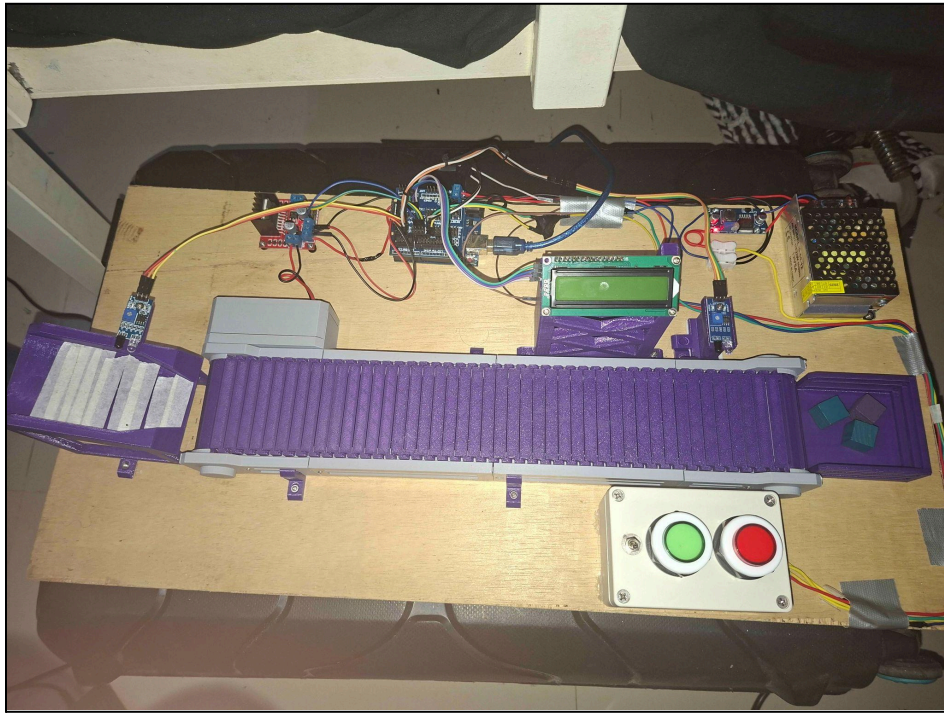
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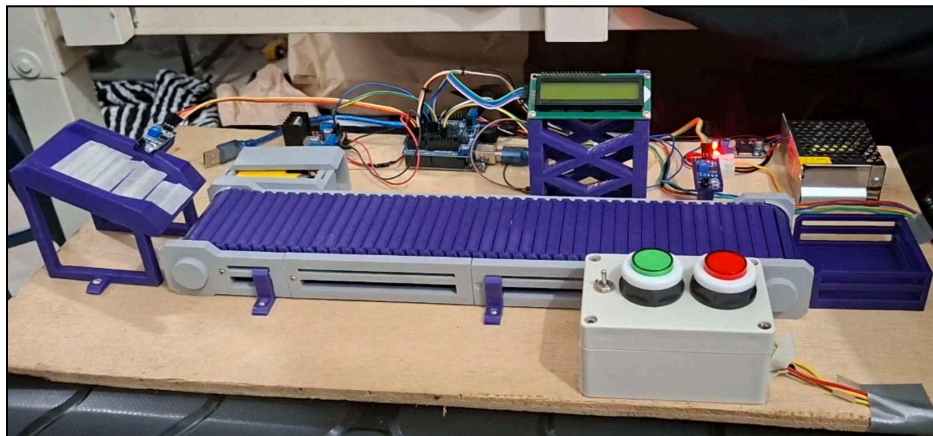
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"top view prototype"



"front view prototype"

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X. References

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